LHAASO Collaboration
Analysis Internal Note

LHAASO Internal Note

Date: July 3, 2026

Search for Ultra-Relativistic Magnetic Monopoles with LHAASO-KM2A

Zhonghua Zhang[1*], Yusheng Wu[1], Ruizhi Yang[2], Yanhong Yu[3]

[1] *Department of Modern Physics, University of Science and Technology of China, Hefei 230026, China*

[2] *Department of Astronomy, University of Science and Technology of China, Hefei 230026, China*

[3] *School of Physics, Henan Normal University, Xinxiang 453007, China*

---

* Main Editor: `zh_zhang@mail.ustc.edu.cn`

Abstract

Extensive air showers induced by ultra-relativistic magnetic monopoles are expected to exhibit a pronounced deficit of muons compared to standard cosmic-ray events. We exploit this distinctive phenomenological signature to search for monopoles using about 4 years of observational data from the LHAASO-KM2A array. By capitalizing on the array's dense, overlapping coverage of electromagnetic and muon detectors, our analysis achieves exceptional discrimination against the vast cosmic-ray background. Finding no evidence of monopole signals, we set stringent 90% confidence level (C.L.) upper limits on the isotropic flux. The derived constraints bound the flux below 1.0×10^{-16} cm^{-2} s^{-1} sr^{-1} at a Lorentz factor $\gamma = 10^5$, and reach 8.2×10^{-19} cm^{-2} s^{-1} sr^{-1} at $\gamma = 10^7$. By exploring previously unconstrained kinematic regimes, this work delivers highly competitive results that strongly complement current bounds from other experimental techniques.

Contents

1 Introduction **3**

2 The LHAASO-KM2A Detector **4**

3 Monte Carlo Simulation: CORSIKA and G4KM2A **5**
- 3.1 Signal Simulation . . . 5
  - 3.1.1 Interaction Models and Mechanisms . . . 5
  - 3.1.2 Signal Statistics and Reference Flux . . . 6
- 3.2 Detector Response Simulation . . . 9

4 Reconstruction and Quality Selection **9**
- 4.1 Standard Reconstruction . . . 9
- 4.2 Energy Reconstruction for Monopoles . . . 9
- 4.3 Basic Reconstructed Quality Selection . . . 13
- 4.4 Justification of the Background Composition Assumption . . . 14

5 Event Selection and Classification **17**
- 5.1 Signal Region Definition . . . 17
- 5.2 Multivariate Analysis (MLP) Strategy . . . 19

6 Data-Driven Background Estimation **21**
- 6.1 Data-Driven Methodology . . . 21
- 6.2 Yield Extrapolation . . . 23

7 Systematic Uncertainties **25**
- 7.1 Background Shape Uncertainty . . . 25
- 7.2 Signal Simulation Uncertainties . . . 27
- 7.3 Summary of Systematics for $\gamma = 10^5$ . . . 28

8 Other Lorentz Phase Space **28**
- 8.1 $\gamma = 10^6$ . . . 28
- 8.2 $\gamma = 10^7$ . . . 30

9 Results **32**
- 9.1 Statistical Framework . . . 32
- 9.2 Upper Limits on Monopole Flux . . . 34
  - 9.2.1 $\gamma = 10^5$ Search . . . 34
  - 9.2.2 $\gamma = 10^6$ Search . . . 36
  - 9.2.3 $\gamma = 10^7$ Search . . . 36
  - 9.2.4 Summary and Physical Interpretation . . . 36
- 9.3 Comparison with Existing Experimental Constraints . . . 38

10 Conclusion **39**

1 Introduction

The existence of magnetic monopoles is one of the most compelling predictions in fundamental physics, originally proposed by P.A.M. Dirac to demonstrate the consistency of quantum mechanics with electromagnetism and to explain the quantization of electric charge [1]. In the framework of Grand Unified Theories (GUTs), 't Hooft and Polyakov independently showed that magnetic monopoles arise inevitably as topological solitons during spontaneous symmetry breaking [2, 3]. While standard superheavy GUT monopoles ($M \sim 10^{16}$ GeV) are expected to be diluted by cosmological inflation to undetectable levels, scenarios involving intermediate symmetry breaking scales or non-standard cosmological histories predict the existence of lighter monopoles that could survive to the present day [4].

This analysis focuses on Intermediate-Mass Monopoles (IMMs) with masses around $M \sim 10^5$ GeV. Such particles are well-motivated within extensions of the Standard Model, such as $SO(10)$ GUTs [5], which allow for a multi-stage symmetry breaking hierarchy: $G \xrightarrow{M_{GUT}} H_1 \xrightarrow{M_I} H_{SM}$. Topological defects generated at the intermediate scale $M_I \sim 10^3 - 10^4$ GeV would possess masses of order $M \sim M_I/\alpha_{gauge} \approx 10^5$ GeV [6]. Unlike their superheavy counterparts, which are likely non-relativistic ($\beta \sim 10^{-3}$), these lighter monopoles can be accelerated by galactic and intergalactic magnetic fields to ultra-relativistic velocities.

From an experimental perspective, a significant unexplored window exists in the monopole phase space. Previous searches have primarily constrained the flux of non-relativistic monopoles (e.g., MACRO [7]) or ultra-high-energy monopoles (e.g., superheavy: IceCube [8]; intermediate mass: Pierre Auger [9], ANITA-II [10] and RICE [11]). Specifically, RICE has set limits for $\gamma \gtrsim 10^7$. However, the regime of ultra-relativistic monopoles with Lorentz factors $10^5 \leq \gamma \leq 10^7$ remains largely unprobed. In this kinematic range, a magnetic monopole with charge $g_D = 68.5e$ behaves as a highly ionizing particle and deposits energy rapidly via ionization, bremsstrahlung, pair production, and photonuclear interactions [12].

The LHAASO-KM2A array, with its high-altitude deployment, vast effective area, and powerful capability to separate the muon component from electromagnetic particles, is uniquely sensitive to the extensive air showers (EAS) initiated by such particles. Furthermore, benefiting from KM2A's exceptional angular resolution, astrophysical gamma-ray background events can be efficiently vetoed by applying an astro-masking technique based on known source catalogs. The shower profile of a monopole in this energy range ($E_{tot} \sim 10^{10} - 10^{12}$ GeV) is expected to differ significantly from that of normal cosmic ray protons. Theoretical simulations indicate that these monopole-induced showers are dominated by intense electromagnetic cascades, resulting in an anomalous muon-poor signature compared to hadronic cosmic ray showers of the same energy scale.

To robustly extract this exotic signal, we developed a dedicated analysis workflow. Given the extreme depletion of Monte Carlo hadronic statistics in the muon-poor phase space, we employ a purely data-driven methodology to model the background, extracting shape and yield templates directly from experimental control regions. A Multi-Layer

Perceptron (MLP) neural network is then utilized to perform multivariate classification, leveraging the topological disparities between the simulated monopole signal and the data-driven background to maximize discrimination power.

In this note, we present a search for ultra-relativistic magnetic monopoles using LHAASO-KM2A data collected from 2021.7 to 2025.7, totally about 1438.81 live days. We describe the simulation of monopole signals, the data-driven background modeling, the development of the MLP classification method, and the evaluation of systematic uncertainties. Finally, we report the flux upper limits established in this previously unexplored kinematic region.

The remainder of this note is organized as follows:

- **Section 2** provides a brief overview of the LHAASO-KM2A detector.
- **Section 3** outlines the Monte Carlo simulation of the monopole signal and proton background.
- **Section 4** describes the application of the hybrid energy reconstruction model optimized for monopole showers, the basic quality selection and the background component.
- **Section 5** describes the definition of signal region (SR) and discriminative variables extracted from shower morphology.
- **Section 6** presents the **Data-Driven Background Estimation** methodology, which provides the background parameter distributions and normalization derived directly from experimental control regions and details the architecture of the Multivariate Analysis (MVA) using a Multi-Layer Perceptron (MLP).
- **Section 7** evaluates the systematic uncertainties, focusing on signal efficiency and the robustness of the data-driven background extrapolation.
- **Section 8** focuses on other two phase space point $10^6, 10^7$.
- **Section 9** reports the final results after unblinding, including the CL_s statistical inference and the 90% confidence level upper limits on the magnetic monopole flux.
- **Section 10** concludes scientific significance of this analysis.

2 The LHAASO-KM2A Detector

The Large High Altitude Air Shower Observatory (LHAASO) is located at Haizi Mountain, Sichuan, China, at an altitude of 4410 m a.s.l. ($\sim$ 600 g/cm^2). The Kilometer Square Array (KM2A) is one of the main components of LHAASO, designed for high-energy γ-ray astronomy and cosmic ray physics[13]. It consists of two types of detectors:

- **Electromagnetic Particle Detectors (EDs):** The array comprises 5216 EDs, each consisting of a 1 m^2 plastic scintillator covered by a 0.5 cm lead plate. EDs are designed to detect the electromagnetic component (electrons, positrons, and photons) of Extensive Air Showers (EAS) with high time resolution, enabling precise reconstruction of the shower arrival direction and core position.
- **Muon Detectors (MDs):** The array includes 1188 MDs, each being a water Cherenkov tank with an active area of 36 m^2 buried under 2.5 m of soil. The soil overburden effectively shields the electromagnetic component, allowing MDs to measure the muon content of the shower. This capability is crucial for distinguishing photon-induced (or monopole-induced) showers from the hadron-dominated cosmic ray background.

3 Monte Carlo Simulation: CORSIKA and G4KM2A

The simulation framework consists of two distinct stages: the extensive air shower (EAS) generation using CORSIKA [14] with CONEX package[15] and the detector response simulation using the GEANT4-based package[16], G4KM2A[13].

3.1 Signal Simulation

Signal samples for ultra-relativistic magnetic monopoles were generated using CORSIKA (v7.7400). In this study, we establish a benchmark scenario with a monopole mass of $M = 10^5$ GeV and a total energy of $E = 10^{10}$ GeV ($\gamma = 10^5$).

3.1.1 Interaction Models and Mechanisms

To accurately simulate the development of the EAS, specific interaction models were selected to handle the particle cascades initiated by the secondary particles:

- **Hadronic Interactions:** High-energy hadronic interactions ($>$ 80 GeV) are treated using the **QGSJET-II-04** model, while low-energy interactions ($<$ 80 GeV) are handled by the **GHEISHA** model.
- **Electromagnetic Interactions:** The electromagnetic component is simulated using the **EGS4** code integrated into CORSIKA.

Monopole Energy Loss Mechanisms: For the monopole primary itself, the simulation accounts for the enormous energy loss driven by its large magnetic charge ($g \approx 68.5e$). The total energy loss rate is dominated by electromagnetic processes, while direct hadronic interactions are explicitly neglected in this model. The specific differential energy loss formulas adopted in the simulation are as follows:

- **Ionization:** Following the modified Bethe-Bloch formula for magnetic charges:

$$\frac{\mathrm{d}E_{\mathrm{coll}}}{\mathrm{d}x} = -\frac{4\pi Z\alpha\alpha_M N_A}{Am_e}\left[\ln\left(\frac{m_e\beta^2\gamma^2}{I}\right) - \frac{\delta}{2}\right] \tag{1}$$

- **Bremsstrahlung:** Given the large mass of the monopole, this contribution is generally suppressed compared to electrons but becomes relevant at ultra-high γ:

$$\frac{\mathrm{d}E_{\mathrm{rad}}}{\mathrm{d}x} \simeq -\frac{16}{3}\alpha Z^2 \alpha_M^2 \frac{N_A}{AM}\gamma \ln \gamma \tag{2}$$

- **Pair Production:** The production of electron-positron pairs is a dominant energy loss channel at high Lorentz factors:

$$\frac{\mathrm{d}E_{\mathrm{pair}}}{\mathrm{d}x} \simeq -\frac{16}{\pi}\alpha^3 \alpha_M \frac{Z^2 N_A}{A m_e}\gamma \left(\frac{M}{m_e}\chi\right) \tag{3}$$

- **Photonuclear Interaction:** The interaction with nuclei via virtual photons scales non-linearly with energy:

$$\frac{\mathrm{d}E_{\mathrm{pho\_nu}}}{\mathrm{d}x} \propto -\gamma^{1.28} \tag{4}$$

In these equations, α is the fine structure constant, α_M is the magnetic fine structure constant, Z and A refer to the medium (air) properties, and M is the monopole mass.

Note on Hadronic Interactions: It is important to note that direct **Hadronic Interactions** for the monopole are **ignored** in this simulation framework. While some theoretical models (e.g., the Rubakov-Callan effect) predict catalysis of nucleon decay, the standard electromagnetic cascade simulation assumes that the cross-section for direct strong interaction between the monopole and air nuclei is negligible compared to the overwhelming electromagnetic energy losses described above. This assumption simplifies the shower generation to a pure superposition of electromagnetic sub-showers along the monopole trajectory.

Shower Characteristics: The combination of these mechanisms leads to a continuous and intense injection of energy along the monopole's path. As illustrated in Figure 1, the ionization dominates at lower γ, while pair production and photonuclear interactions take over at higher energies. Quantitatively, the resulting EAS for a 10^{10} GeV monopole exhibits a shower size comparable to that of a standard cosmic ray proton with an energy of approximately 160 TeV, but with a significantly distinct longitudinal profile.

3.1.2 Signal Statistics and Reference Flux

The key configuration parameters used in the CORSIKA steering file for the benchmark simulation ($E = 10^{10}$ GeV) are summarized in Table 1 and signal simulation statistics is shown in Table.2.

To evaluate the sensitivity of the LHAASO-KM2A detector to magnetic monopoles, we define a reference signal flux based on the **Parker Bound**, which represents the

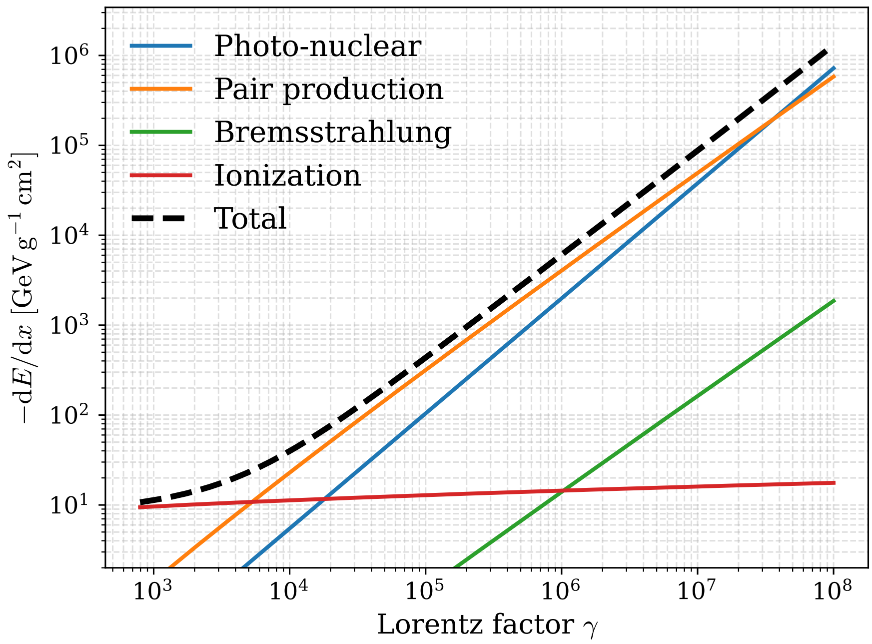


Figure 1: **Differential energy loss of a magnetic monopole in air as a function of the Lorentz factor γ.** The plot shows the contributions from ionization (collisional), pair production, bremsstrahlung, and photonuclear interactions. For the benchmark case of $\gamma = 10^5$, pair production is the dominant mechanism.

Table 1: Key configuration parameters for the CORSIKA simulation (Example for $E = 10^{10}$ GeV).

Parameter	**Value**	**Description**
PRMPAR	43	Primary particle (43 = Monopole in CONEX)
ESLOPE	0	Slope of primary energy spectrum
ERANGE	1.E10 1.E10	Energy range of primary particle (GeV)
THETAP	0. 70.	Range of zenith angle (degree)
PHIP	-180. 180.	Range of azimuth angle (degree)
OBSLEV	4410.E2	Observation level (4410 m in cm)
FIXHEI	0. 0.	Starting altitude (g/cm^2)
MAGNET	20.0 42.8	Magnetic field components (B_x, B_z in μT)
THIN	1.E-12 1.E0 0.E0	Thinning parameters
ECUTS	0.1 0.1 1.E-3 1.E-3	Energy cuts (GeV)
CONEX	1.E-3 1 1.E-7	Energy thresholds for cascade equations

Table 2: **Signal** simulation statistics for Full-KM2A

Energy Range	Simulated Events
10^{10} GeV	2.52×10^5
10^{11} GeV	1.00×10^4
10^{12} GeV	5.40×10^3

theoretical upper limit imposed by the survival of the Galactic magnetic field. For the mass range of interest ($M \sim 10^5$ GeV), the Parker Bound is approximately:

$$\Phi_{\text{Parker}} \sim 10^{-15}\,\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1} \tag{5}$$

Assuming a hypothetical flux exactly equal to the Parker Bound ($\Phi = \Phi_{\text{Parker}}$)[17], we calculate the total number of expected monopole events (N_{ref}) injected into the simulation volume over a standard observation period of one year. The geometric acceptance is defined by a circular area with a radius of $R = 1000$ m (covering the full KM2A array) and a zenith angle range of $0° \leq \theta \leq 70°$.

The variation of the detector's geometric exposure with zenith angle, after applying the astrophysical masking (excluding the Galactic plane and bright sources), is shown in Figure 2. Integrating this exposure over one year, the total expected number of signal events corresponding to the Parker Bound is calculated to be approximately 2941 (calculated as $\Phi \cdot A \cdot \Omega \cdot T$). This value serves as the normalization factor for determining the final flux upper limit.

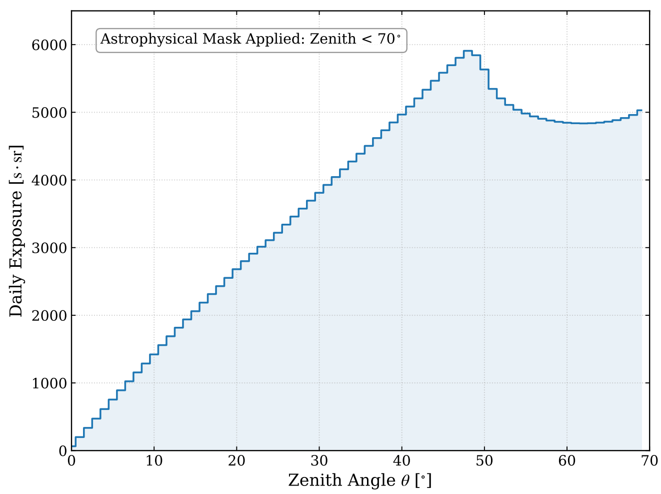


Figure 2: **Geometric exposure of the LHAASO-KM2A array as a function of zenith angle.** The distribution accounts for the astrophysical masking (Galactic plane and point sources) shown in Fig.6 applied in the analysis. The drop in exposure at specific angles reflects the geometric exclusion of these sky regions.

Table 3 summarizes the signal survival efficiency at various stages of the analysis chain. Starting from the generated reference number, we list the remaining events after the trigger, basic quality cuts, and the final MVA selection. Detailed description is listed in Section 4 and 5

Table 3: 10^{10} **GeV Signal reduction flow for a reference flux at the Parker Bound during 1 year.** The "Efficiency" column represents the cumulative efficiency relative to the generated number (N_{gen}). The final row indicates the expected number of signal events in the signal region if the monopole flux were equal to $10^{-15}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$.

Selection Stage	**No. of Events (N)**	**Relative Efficiency (%)**
Generated (N_{gen})	2941 (Ref. Value)	100.00 %
Triggered (N_{trig})	1563	53.12 %
Quality Cuts (N_{qual})	944	60.44 %
Signal Region (N_{SR})	296	31.32 %

3.2 Detector Response Simulation

The particle lists generated by CORSIKA are passed to the detector simulation package **G4KM2A** (based on GEANT4). This step simulates the interaction of secondary particles with the KM2A detector components, including the scintillator response in EDs and the Cherenkov light production in MDs.

The key input parameters for the G4KM2A simulation are:

- **ArrayMode:** Set to `FullArray` (representing the complete KM2A configuration).
- **AreaRadius:** For the benchmark 10^{10} GeV signal, the radius is set to `1000 m` to match the CORSIKA scatter area and ensure consistent edge effect handling. It should be noted that for other energy points, this radius is slightly adjusted to optimize simulation efficiency while maintaining sufficient coverage of the array's effective area.

4 Reconstruction and Quality Selection

4.1 Standard Reconstruction

Using G4KM2A reconstruction code version V3.

4.2 Energy Reconstruction for Monopoles

Standard energy reconstruction for LHAASO-KM2A is primarily optimized for transient showers initiated by γ rays or cosmic ray nuclei. In typical γ-ray analyses, the

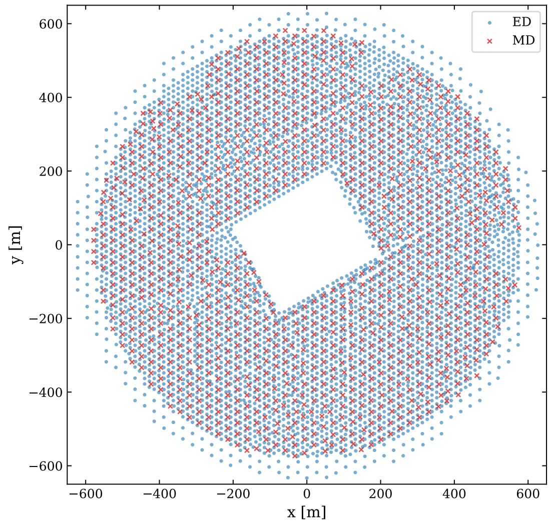


Figure 3: Layout of KM2A. The blue dots represent the 5216 ED units, while the red crosses denote the 1188 MD units.

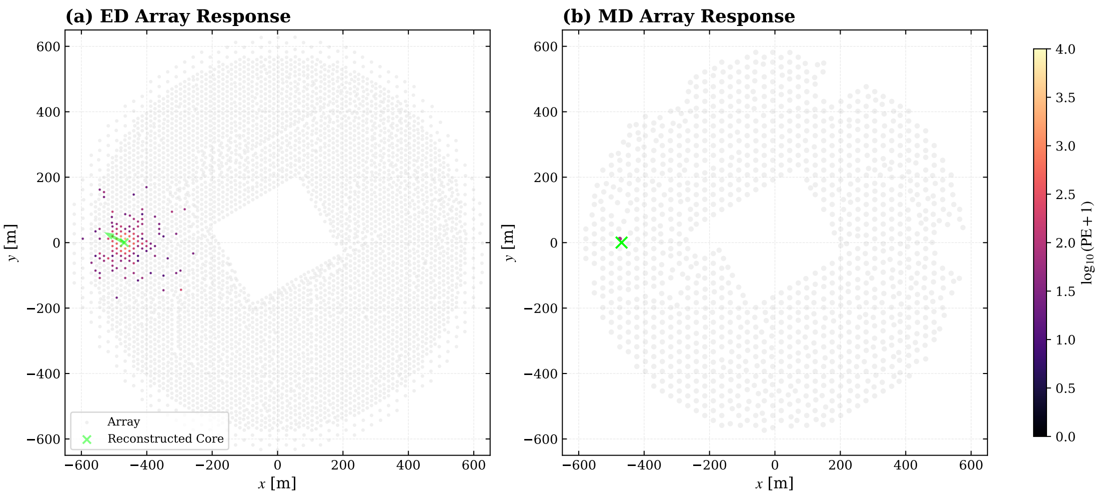


Figure 4: Display of a typical shower event induced by magnetic monopole with $E = 10^{10}$ GeV and $M = 10^5$ GeV. (a) ED hits with reconstructed core (green cross) and arrival direction (green arrow). (b) Corresponding MD hits.The only unit active in MD indicates a special feature of monopole shower: less muon produced . The color scale indicates the photoelectron (PE) yield. Gray dots represent the inactive detector units in this event.

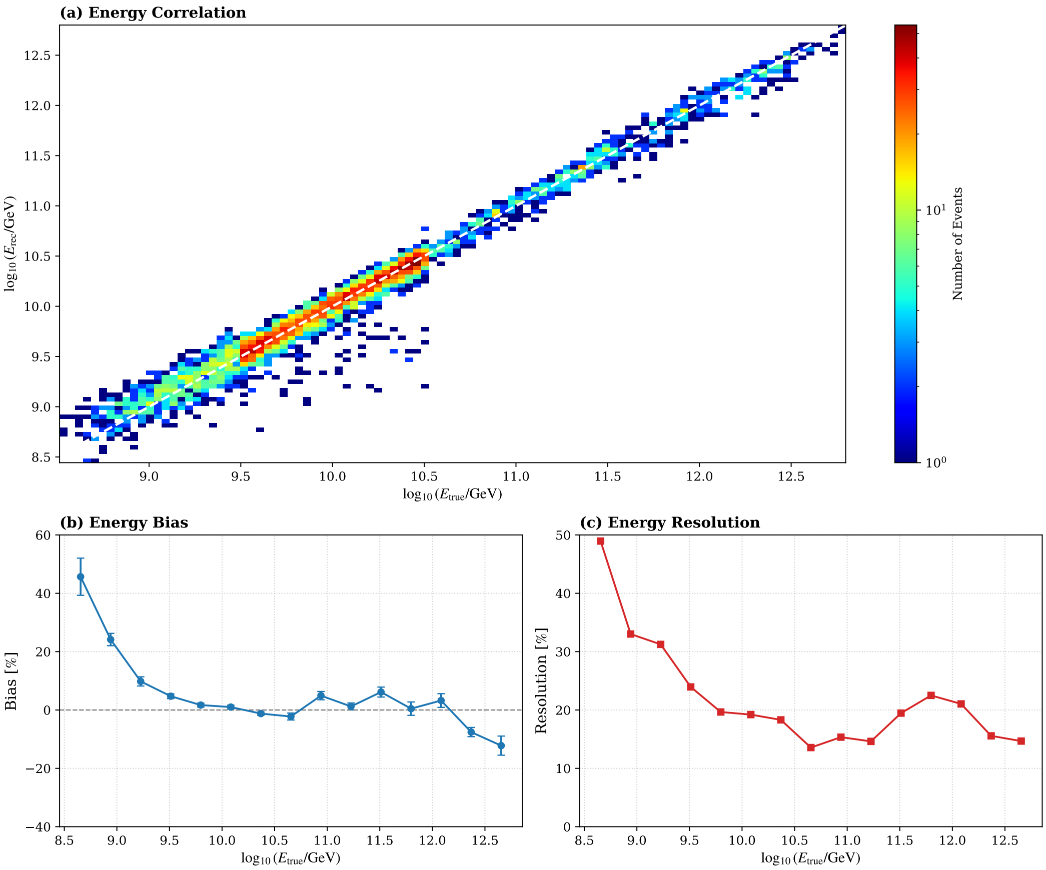


Figure 5: **Performance of the GBDT-based energy reconstruction for high-energy particles.** (a) Correlation between the reconstructed energy ($\log_{10} E_{rec}$) and the true primary energy ($\log_{10} E_{true}$). (b) Energy bias as a function of primary energy: $\langle \frac{E_{rec}-E_{true}}{E_{true}} \rangle$. (c) Energy resolution (%) across the simulated energy range: $\sigma(\frac{E_{rec}-E_{true}}{E_{true}})$. The results demonstrate that the reconstruction is stable and sufficient for defining the energy scale in this analysis.

reconstructed energy relies on the shower age, size, and zenith angle, strictly requiring the reconstructed zenith angle to be less than 50°. However, in the search for magnetic monopoles, we extend the zenith angle acceptance to 70° to maximize the geometric exposure. Furthermore, the energy scale and energy loss mechanisms of ultra-relativistic monopoles differ significantly from those of standard cosmic rays. Therefore, a dedicated energy reconstruction method is required.

To achieve robust and precise energy reconstruction across a vastly extended energy range, we developed a hybrid regression model combining a multiple linear baseline with a Gradient Boosting Decision Tree (GBDT, implemented via XGBoost[18]). While tree-based models are exceptionally powerful at capturing complex, non-linear correlations, they inherently lack extrapolation capabilities. For inputs falling outside the training phase space, a standard GBDT maps predictions to the nearest leaf node boundary, resulting in an unphysical "flattening" of the reconstructed energy. Given the limited statistics of Monte Carlo simulations at the extreme high-energy frontier, relying solely on a GBDT would introduce severe boundary biases.

To overcome this limitation, we employ a residual learning strategy. The reconstructed energy is formulated as the sum of a linear component and a non-linear residual component:

$$\log_{10}(E_{rec}) = f_{\text{linear}}(X) + f_{\text{XGB}}(X) \tag{6}$$

where X represents the vector of input features. The primary input features include the reconstructed zenith angle (θ_{rec}) and the particle density at a lateral distance of 50 m from the shower core (ρ_{50}^{NKG}), which is derived from the detector-level NKG fit [19].

In this architecture, the linear regression model (f_{linear}) captures the global physical trends—such as the roughly linear logarithmic correlation between shower size and primary energy, modulated by atmospheric attenuation at different zenith angles. This ensures a natural and physically sound extrapolation into the highest energy regimes where training data is sparse. Subsequently, the XGBoost model (f_{XGB}) is trained strictly on the residuals (the difference between the true energy and the linear prediction). This component handles local non-linearities and detector response nuances only in the phase space where statistical support is sufficient, effectively tapering off outside the training boundaries without disrupting the linear extrapolation.

The performance of this hybrid model is evaluated using independent Monte Carlo samples. The inclusion of the linear baseline fundamentally resolves the high-energy saturation issue. Quantitative tests demonstrate that the pure linear model already establishes a strong correlation ($R^2 \approx 0.977$), while the addition of the XGBoost residual correction further minimizes the mean absolute error (MAE) and refines the overall fit ($R^2 \approx 0.979$). As shown in Figure 5, the correlation between the reconstructed and true primary energy exhibits excellent linearity across the entire energy range of interest, maintaining a stable energy bias and an energy resolution of approximately 20% above 10^{10} GeV.

It should be emphasized that for the purpose of this magnetic monopole search, high-precision energy spectroscopy is not the primary objective. Instead, the energy reconstruction is utilized to establish a consistent energy scale that allows for the robust selection of cosmic ray background events with an Extensive Air Shower (EAS) scale comparable to that of the signal. For instance, this methodology allows us to effectively isolate the ~ 160 TeV proton-induced showers that morphologically mimic the 10^{10} GeV monopole signal, ensuring that the subsequent multivariate discrimination is performed on a physically relevant phase space.

4.3 Basic Reconstructed Quality Selection

Before performing signal-background discrimination, a set of preliminary quality cuts is applied to both experimental data and Monte Carlo samples. The aim is to ensure that the reconstructed shower parameters—such as core position, direction, age and energy—are reliable.

The basic quality selection criteria include:

- **Hit and Photo-electron Statistics:**
  - **Triggered Units (N_{trigE}):** A minimum of 19 triggered ED units is required to ensure a stable event trigger.
  - **Filtered Hits (N_{filtE}, N_{hitM}):** We require $N_{filtE} \geq 10$ for EDs and $N_{hitM} \geq 10$ for MDs to provide sufficient data points for lateral distribution fitting.
  - **Photo-electron Count ($NpE3$):** The total number of photo-electrons(PEs) within 400 m of the core must be ≥ 250 to ensure the event possesses enough energy for a valid GBDT reconstruction. The threshold is selected based on the observation that most of monopole shower with $E = 10^{10}$ GeV have NpE3 more than 250. So we can safely ignore events which have less PEs in other cases.
  - NpE **Ratio** A lateral distribution ratio criterion is applied ($NpE1/NpE2 \geq 2.0$, where $NpE2 > 0$). This ensures the shower exhibits a well-defined, compact core.
  - **Muon Validity ($NuM1$):** We require the muon count within 200 m ($NuM1$) to be non-negative ($NuM1 \geq 0$), serving as a basic validity check for the MD reconstruction.
- **Geometric, Angular, and Sky Acceptance:**
  - **Core Position (dr):** To maximize the expected exposure while maintaining reconstruction quality, the shower core must be located within the range of detector arrays, typically defined as $dr > 0$.
  - **Zenith Angle (θ_{rec}):** Consistent with our objective to maximize exposure, the reconstructed zenith angle is restricted to $0° \leq \theta_{rec} \leq 70°$.
  - **Astrophysical Masking (Data Only):** To eliminate contamination from high-energy γ-ray sources, which also exhibit a "muon-poor" signature similar to magnetic monopoles, we apply a geometric mask to the experimental data:

* **Galactic Plane:** Events located within the Galactic latitude band of $|b| < 10°$ are excluded to reject the diffuse γ-ray emission from the Galactic disk.
* **Known Sources[20]:** A spatial veto is applied around known significant Very High Energy (VHE) γ-ray sources (e.g., standard candles like the Crab Nebula and bright LHAASO sources) to prevent point-source mimicking.

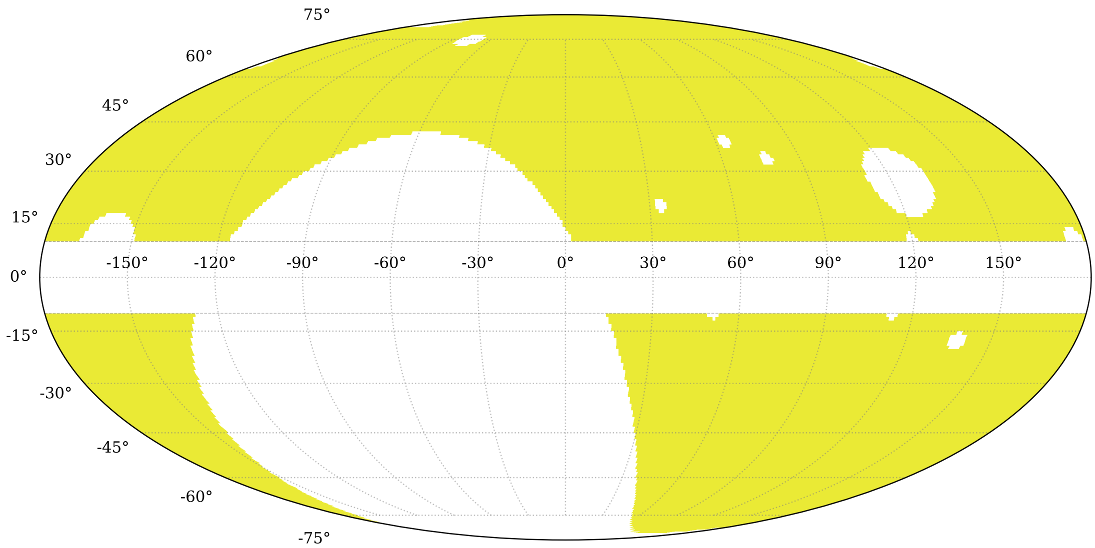


Figure 6: **The geometric mask applied to experimental data in equatorial coordinates.** The yellow region represents the effective acceptance area for this analysis. The white regions indicate the masked areas, including: (1) The Galactic Plane band ($|b| < 10°$), masked to reject diffuse γ-ray emission and sources in it; (2) Discrete circular regions masking known high-energy γ-ray sources (e.g., Crab, LHAASO J1908, etc.) to prevent point-source contamination. This masking ensures that the "muon-poor" dataset used for the monopole search is not contaminated by astrophysical γ-ray signals.

4.4 Justification of the Background Composition Assumption

In the simulation strategy described in Section 3, the cosmic ray background is modeled exclusively using proton samples. This simplification is adopted not only for computational efficiency but also because it provides a conservative estimate of the background contamination in the signal region.

The rationale relies on the observation that protons constitute the "irreducible" background closest to the monopole signal characteristics. While simulating the full cosmic ray flux (including He, CNO, MgAlSi, Fe) across the entire phase space ($0-1000$ m, $0-70°$) is computationally prohibitive, we validated this assumption using a ready-made dataset generated in a restricted geometry ($260-480$ m, $0-40°$).

Figure 7 presents the comparison of discrimination variables for different cosmic ray components within reconstructed energy window of $E = 10^{10}$ GeV monopole shown in Equ. 7. The implications for our analysis are as follows:

1. **Muon Content Differences:** As shown in Figure 7(a), the muon content scales with the atomic mass number A. Heavier nuclei (e.g., Iron) generate significantly more muons than protons. Since the monopole signal is defined by a "muon-poor" signature which shown in Fig.10, the proton component sits at the "leftmost" boundary of the hadronic background distribution, making it the most "signal-like" among all cosmic ray species.

2. **Preferential Suppression of Heavy Nuclei:** As we tighten the selection cuts (specifically the $R_{\mu e}$ and shower age cuts) to approach the Signal Region (SR), the suppression efficiency for heavier nuclei is significantly higher than that for protons. In the deep signal-like region, the surviving fraction of heavy nuclei becomes negligible compared to protons.

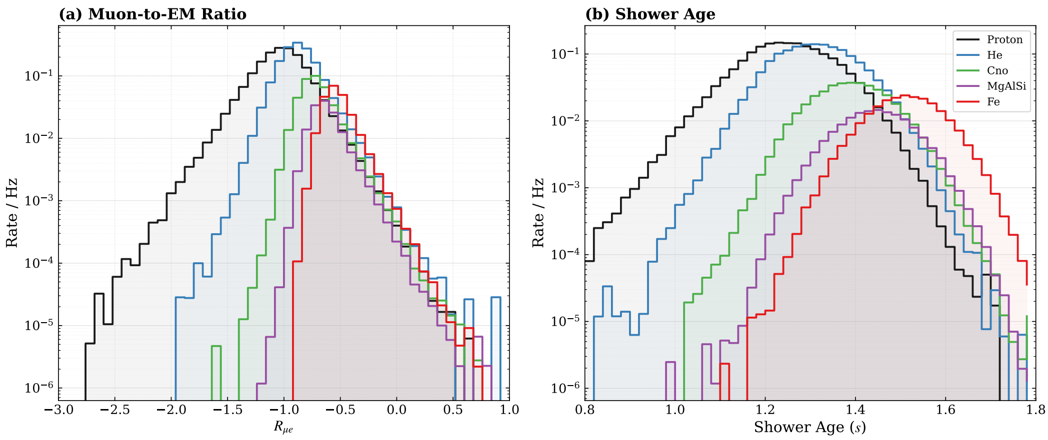


Figure 7: **Distributions of discrimination variables for different cosmic ray components.** (a) $R_{\mu e}$ and (b) Shower Age distributions from the validation dataset. The plots demonstrate that protons (black) are the closest to the signal region (low $R_{\mu e}$). As cuts are applied to isolate the signal, heavier components (colored lines) are suppressed more rapidly, leaving protons as the dominant and most challenging background source.The rate is number of different component per second captured by selected area and zenith angle in KM2A

Also, Figure 8 strongly indicates that in the potential signal region, the number ratio of proton to helium which is the second most in muon-poor region, is high enough to support the component assumption.

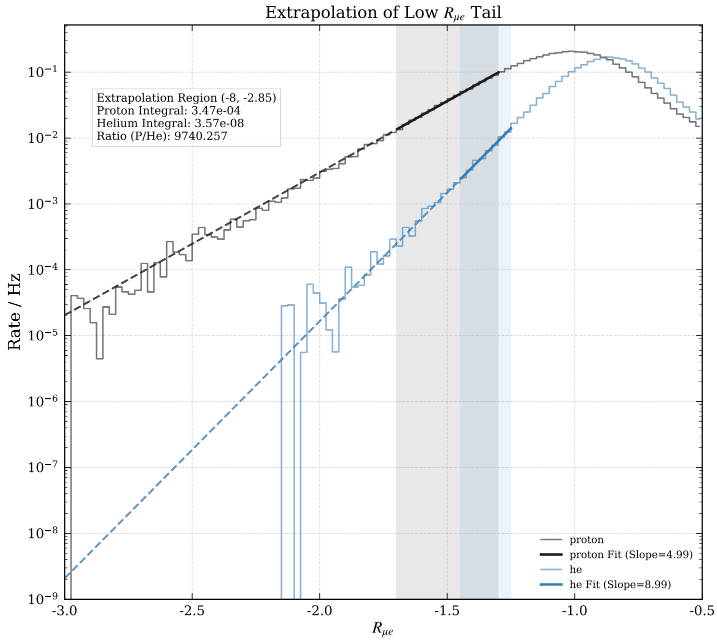


Figure 8: $R_{\mu e}$ **parameter extrapolation of proton and helium.** In log scale y-axis, the rate of proton and helium increase linearly as $R_{\mu e}$. To validate the component assumption of only considering proton as background quantitatively, we extrapolate the linear relationship to range [-8.0, -2.84] (largely covering signal region) and integrate the rate in it. The ratio of proton and helium is approximately 1.6×10^4 which strongly supports the component assumption.

Conclusion: By modeling the background exclusively as protons, we effectively treat any potential surviving heavy nuclei in the data as if they were protons. Since protons are inherently harder to distinguish from monopoles than heavier nuclei, this assumption treats the remaining background as "more signal-like" than it might actually be. This leads to a **conservative estimation**: if our analysis can effectively reject the proton background, it will, by definition, reject the heavier components with even greater margin. Thus, the proton-only model is a safe and robust proxy for the total cosmic ray background in this search.

5 Event Selection and Classification

Following the basic quality pre-selection described in Section 4.3, we proceed to extract the physical features required to distinguish the magnetic monopole signal from the standard cosmic-ray background. A fundamental challenge in this rare-event search is the severe depletion of Monte Carlo (MC) background statistics in the extreme muon-poor phase space. As illustrated in the E_{rec} vs. $R_{\mu e}$ distribution (Figure 10), applying the strict $R_{\mu e}$ cuts required to approach the Signal Region (SR) leaves virtually no surviving events in the MC proton sample. Consequently, relying on simulated hadronic showers to model the background deep in the less-muon tail is both statistically unfeasible and systematically unreliable. To overcome this limitation, we entirely abandon the MC background and adopt a purely data-driven methodology, utilizing experimental data from an adjacent Control Region (CR) to construct our background templates.

For the final signal discrimination, a Multi-Layer Perceptron (MLP) neural network [21] is employed to maximize the multidimensional separation power. The MLP is trained using a hybrid dataset: the simulated monopole sample serves as the signal, while the data-driven CR sample serves as the background. We rigorously evaluate the network's performance to ensure valid generalization without overtraining.

The unified classification score generated by the MLP serves as the nominal observable for the final statistical inference. Furthermore, systematic fluctuations in this score distribution—stemming from background shape extrapolation envelopes, signal energy calibration uncertainties, and selection efficiencies—are directly parameterized and incorporated as inputs into the `RooStats` workspace for the upper limit estimation (detailed in Sections 7 and 9).

Note that to provide a clear and consistent demonstration of the analysis workflow, the distributions and methodology presented in the main body of this section will focus on the benchmark kinematic point: an ultra-relativistic monopole with a Lorentz factor of $\gamma = 10^5$ and a primary energy of $E = 10^{10}$ GeV.

The overview of the analysis is figured in Figure 9

5.1 Signal Region Definition

A rough estimation of the background rate—dominated by cosmic ray protons—suggests that even if applying the selection of $R_{\mu e} < -1.95$, the remaining background count could

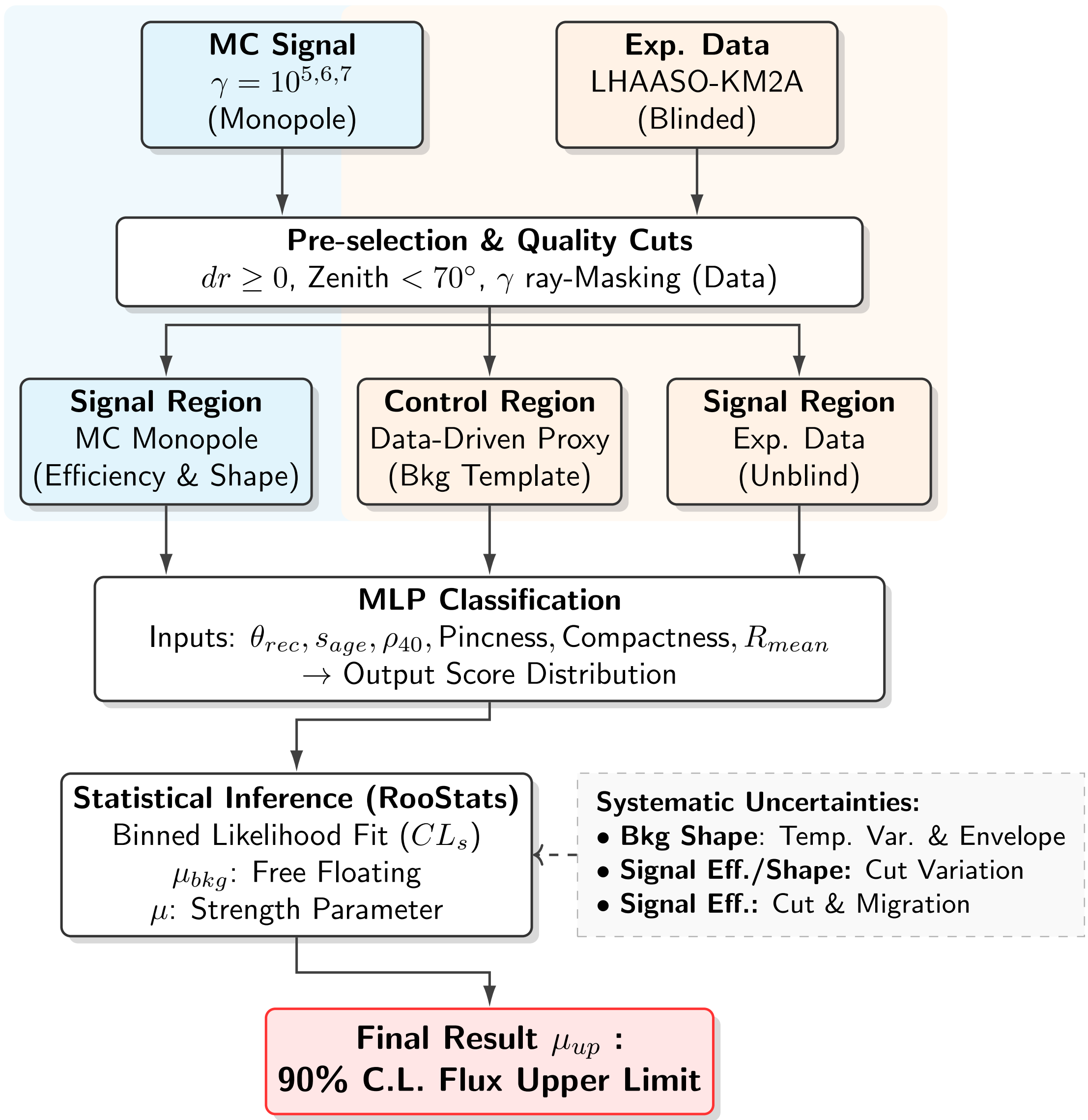


Figure 9: **Analysis Workflow.** The flowchart outlines the procedure from data/simulation preparation, through pre-selection and MLP classification, to the final statistical inference. The background template is derived using a data-driven approach from the experimental control region.

reach approximately 2×10^6 events under exposure of one year. In stark contrast, the expected number of monopole signals surviving the same cut is only around 500. This overwhelming background dominance, surpassing the signal by four orders of magnitude, far exceeds the separation capabilities of any standard analysis. Therefore, a more stringent set of selection cuts is imperative to isolate the signal candidates.

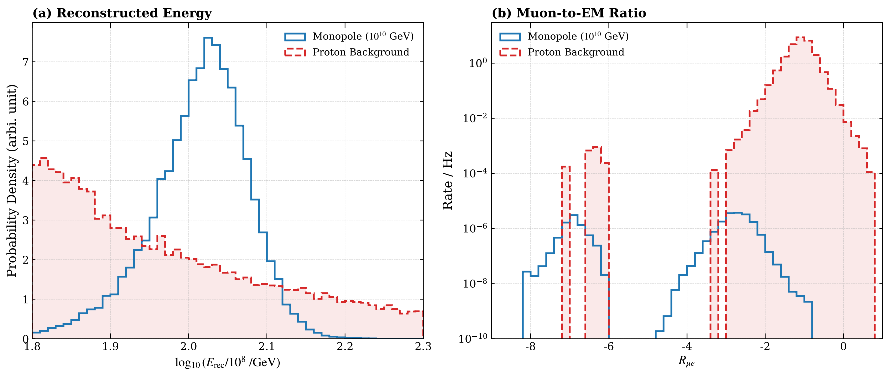


Figure 10: **Distributions of key discrimination variables for the benchmark channel (10^{10} GeV).** (a) The probability density of the reconstructed energy. The monopole signal forms a peak distinct from the power-law cosmic ray background. (b) The rate distribution of the muon-to-electron ratio $R_{\mu e}$. The electromagnetic nature of the monopole shower results in a "muon-poor" signature, well-separated from the hadronic background. And the MC proton here is just used for exhibition, real background template is from data driven in Section.

- **Reconstructed Energy Window (E_{rec}):** To focus on the energy range relevant to the signal, we select events within a specific window around the expected peak. For the 10^{10} GeV benchmark, this is defined as:

$$1.95 < \log_{10}(E_{rec}/10^8 \text{ GeV}) < 2.10 \tag{7}$$

  This corresponds to the signal-dominated peak shown in Fig. 10(a).

- **Muon-to-Electron Ratio ($R_{\mu e}$):** This remains the most powerful discriminator. A strict cut of $R_{\mu e} < -2.90$ is applied to maximally suppress the hadronic background and retain only the most "muon-poor" candidates.

5.2 Multivariate Analysis (MLP) Strategy

Based on the distinct shower characteristics quantified by some parameters shown in Figure.12, we apply the following selection criteria:

- **Shower Morphology Parameters:** In addition to energy and muon content, we utilize auxiliary variables describing the lateral structure and smoothness of the shower to reject anomalous events or reconstruction artifacts.

- **Shower Age (s_{age}):** We constrain the lateral age parameter to a physically reasonable range of $0.8 < s_{age} < 1.6$. This range effectively filters out numerical fitting errors and non-physical events while preserving the majority of the monopole signal, which typically exhibits a stable age profile.
- **Lateral Density (ρ_{40}):** Defined as the average particle density for hits with a lateral distance $r > 40$ m ($\rho_{40} = \sum_{r>40} PE/N_{r>40}$). We require $\rho_{40} \in [40, 150]$ to select high-energy cores with significant lateral spread, distinguishing them from low-energy local fluctuations.
- **Pincness:** A metric quantifying the smoothness of the lateral distribution, calculated as the variance of hit intensities within radial bins (normalized by local mean). A cut of $0.5 <$ Pincness < 1.0 is applied. Monopole showers, being driven by continuous electromagnetic processes, tend to have smoother lateral profiles compared to the "clumpy" nature of hadronic sub-showers, falling well within this range.
- **Compactness:** Defined as the ratio of the total number of hits to the maximum PE recorded beyond 45 m ($N_{hits}/\max(PE)_{r>45}$). A range of $0 <$ Compactness < 1.0 is selected to ensure the event has a dense core relative to its periphery, characteristic of the intense ionization of a monopole.
- R_{mean}**:** Defined as the average of radius of hits from reconstructed core: $\frac{\Sigma PE \times r}{\Sigma PE}$. And based on the distribution of monopole, we will set a range to constrain it and make sure it would not disturb the training of classifier in the next subsection.

After applying the basic quality selections, the next critical step is to maximize the separation power between the rare magnetic monopole signal and the overwhelming cosmic-ray background. Because the shower characteristics of monopoles and standard cosmic rays exhibit complex, non-linear correlations, a Multivariate Analysis (MVA) approach is highly preferred over simple, sequential one-dimensional cuts. For this binary classification task, we implement a Multi-Layer Perceptron (MLP) neural network.

The input features fed into the network consist of the six key shower morphological and kinematic parameters: θ_{rec}, s_{age}, ρ_{40}, Pincness, Compactness, and R_{mean}. To ensure the network learns the true physical background rather than potential simulation artifacts, the MLP is trained using a hybrid dataset. The signal class is represented by the simulated monopole Monte Carlo sample, while the background class is populated using experimental data selected from a rigorously defined, signal-depleted Control Region. The precise definition, theoretical justification, and validation of this data-driven background sample will be detailed subsequently in Section 6.

Given the distinct topological separation already inherent in the chosen physical parameters, a moderately complex network architecture is sufficient to achieve optimal discrimination without introducing unnecessary computational overhead. The detailed architecture of the MLP is summarized in Table 4. The network features an input layer matching the dimension of the feature space, two fully connected hidden layers (with 64 and 32 neurons, respectively), and a single output neuron. To mitigate overfitting at the

Table 4: **Architecture of the MLP used for signal discrimination.** The model incorporates Batch Normalization and Dropout layers to enhance generalization and prevent overfitting during the training on the hybrid data-simulation sample.

Layer Stage	Layer Components	Parameters / Settings
Input Layer	Linear	Input Dimension = $N_{features}$
Hidden Layer 1	Linear	Nodes = 64
	Activation	ReLU
	Normalization	BatchNorm1d
	Regularization	Dropout ($p = 0.2$)
Hidden Layer 2	Linear	Nodes = 32
	Activation	ReLU
	Normalization	BatchNorm1d
	Regularization	Dropout ($p = 0.2$)
Output Layer	Linear	Nodes = 1
	Activation	Sigmoid

architectural level, Batch Normalization and Dropout operations are applied after each hidden layer.

Furthermore, to maximize the utilization of the available statistics and rigorously ensure the model generalizes robustly to unseen data, the training procedure employs a k-fold cross-validation strategy [22]. By partitioning the dataset into k mutually exclusive subsets, this technique guarantees that every event is evaluated as an independent test sample exactly once while being shielded from the training phase of its respective fold. This approach systematically suppresses potential selection biases, provides a highly stable evaluation of the classifier's performance, and mitigates the risk of overtuning to specific statistical fluctuations in the hybrid training sample.

The final output, as shown in Figure 14, is transformed by a Sigmoid activation function, effectively mapping the multidimensional input phase space into a single continuous classification score ranging from 0 (highly background-like) to 1 (highly signal-like). This unified score simplifies the complex variable space and serves as the primary observable for the final statistical likelihood fit.

6 Data-Driven Background Estimation

6.1 Data-Driven Methodology

Due to the limited statistical power of the Monte Carlo (MC) simulations in the extreme muon-poor phase space, we employ a purely data-driven method to estimate both the background yield and the probability density functions (PDFs) of the multivariate input parameters within the Signal Region (SR). As defined in Sec. 5.1, the SR is constrained by $1.95 \leq E_{rec} \leq 2.10$ and $-5.00 \leq R_{\mu e} \leq -2.90$.

To achieve this, we define a background-dominated Control Region (CR) consisting of experimental data immediately adjacent to the SR. Utilizing a data-driven CR offers a distinct advantage: it inherently encapsulates the true physical detector response and the actual cosmic-ray composition. Consequently, this approach effectively eliminates the systematic uncertainties associated with theoretical hadronic interaction models and bypasses the need to assume specific primary compositions (e.g., proton, helium, or heavier nuclei fractions).

The fundamental premise of extracting the SR background template from the CR rests on the hypothesis of *shape invariance* in the deep muon-poor tail. We postulate that for highly anomalous extensive air showers, the kinematic and morphological distributions decouple from the $R_{\mu e}$ variable once a sufficiently tight threshold is surpassed.

To justify this hypothesis, we first examine the macroscopic evolution of shower parameters across a broad range of $R_{\mu e}$. Among the discrimination variables, the shower age (s_{age}) exhibits the strongest correlation with the muon content, making it an ideal probe for this transition. As illustrated in Figure 11, the normalized s_{age} distribution demonstrates a clear two-stage behavior in the experimental data. In the muon-rich and intermediate regimes (dotted lines, $R_{\mu e} > -1.5$), the distribution shifts significantly towards younger shower ages as the muon content decreases. However, as the selection enters the muon-poor domain (solid lines, $R_{\mu e} < -1.5$), this dependency abruptly vanishes. The distributions converge, and the shape stabilizes remarkably. This observation confirms that the shower morphology decouples from $R_{\mu e}$ beyond this threshold.

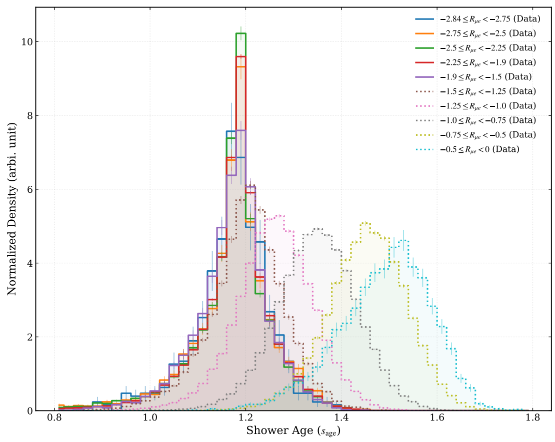


Figure 11: **Validation of background shape stability using experimental data.** The plot shows the normalized distributions of Shower Age (s_{age}) in data for different $R_{\mu e}$ intervals. While the distribution shifts significantly in the muon-rich regions (dotted lines), it stabilizes remarkably for $R_{\mu e} < -1.5$ (solid lines). Since s_{age} is the variable most sensitive to $R_{\mu e}$, this stability justifies using the $R_{\mu e} < -1.5$ sample to approximate the background shape in the signal region ($R_{\mu e} < -2.84$).

Building upon this macroscopic observation, we further verified the shape invariance for all input features specifically in the immediate vicinity of the SR boundary. We examined the variable distributions across successive narrow slices of $R_{\mu e}$ approaching the SR (e.g., from -2.65 down to -2.85, while keeping $1.95 \leq E_{rec} \leq 2.10$ fixed). As illustrated in Figure 12, the parameter distributions demonstrate exceptional stability; the shape of the background remains completely consistent across these varied thresholds deep in the CR. This strict stabilization robustly validates our methodology, allowing us to confidently use the high-statistics data from the adjacent CR as a reliable proxy for the background shape within the SR.

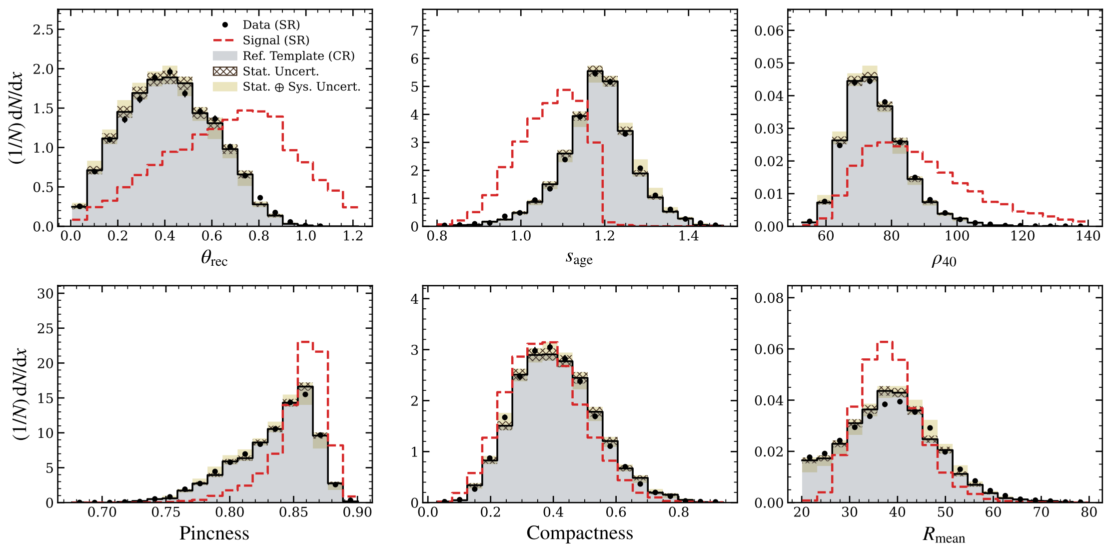


Figure 12: **Validation of background shape invariance and unblinded Signal Region data.** The normalized distributions of key discrimination variables (e.g., shower age, zenith angle, compactness) are shown for various Control Region (CR) slices adjacent to the Signal Region (solid colored lines). The unblinded experimental data within the explicitly defined continuous Signal Region ($-5.0 < R_{\mu e} < -2.9$) is overlaid as the shaded histogram. The excellent shape agreement between the unblinded SR data and the adjacent CR slices provides definitive validation of the shape invariance hypothesis, confirming that the data-driven proxy perfectly models the true background. The expected distribution for a simulated magnetic monopole signal (dashed blue line) is superimposed to highlight the discrimination power of these features.

6.2 Yield Extrapolation

Having established the shape stability of the background variables in the Control Region (CR), we must also establish a baseline expectation for the absolute background yield in the Signal Region (SR). The distribution of $R_{\mu e}$ exhibits a continuous "muon-poor" tail alongside a discrete "zero-muon" peak. The latter originates from events that fail to trigger any muon detectors, which are artificially separated into a distinct peak around $R_{\mu e} < -5.0$ due to the regularization term in $R_{\mu e} = \log_{10}((10^{-4} + N_\mu)/N_{em})$. Because these zero-muon events demonstrate topological and kinematic features that are fundamentally distinct from the continuous less-muon showers, incorporating them would compromise the robustness and consistency of the background modeling. Consequently, we explicitly veto the zero-muon component by restricting our analysis phase space to $R_{\mu e} > -5.0$, focusing our background estimation solely on the continuous muon-poor tail.

For this continuous tail, the $R_{\mu e}$ distribution exhibits a clear exponential decay, which appears linear under a logarithmic scale. We define a sideband fitting region adjacent to the SR (e.g., $-2.8 \leq R_{\mu e} \leq -2.0$) and perform a log-linear extrapolation past the SR boundary ($R_{\mu e} \leq -2.90$). As shown in Figure 13, this provides a baseline integral for the expected background yield. To validate the robustness of this method, an identical sideband extrapolation was tested in a neighboring energy validation region ($1.8 \leq E_{rec} \leq 1.95$), revealing a methodological bias of approximately 34%.

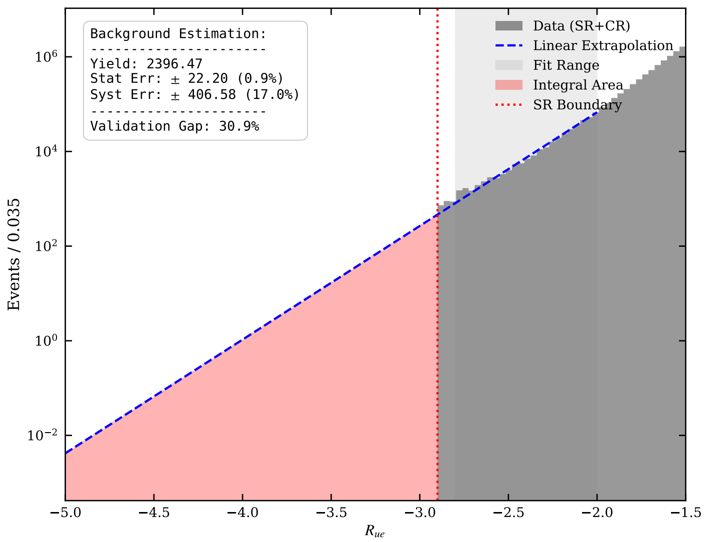


Figure 13: **Log-linear extrapolation of the continuous muon-poor tail.** The background event density is fitted in the sideband region and extrapolated past the SR boundary (red dotted line at $R_{\mu e} = -2.90$). The shaded red area represents the integrated yield for this continuous background component. Note that the event yields shown in this figure are scaled to an equivalent exposure of one year.

It is crucial to emphasize that while this sideband extrapolation provides a vital order-of-magnitude validation for our data-driven model, the explicitly calculated yield and its associated extrapolation uncertainties are **not** rigidly enforced as prior constraints in the final statistical inference. Because sideband extrapolations inherently carry statistical fluctuations and methodological biases (such as the aforementioned 34%

non-closure), treating this extrapolated yield as an absolute truth could artificially over-constrain the final limit.

Instead, during the final likelihood fit, we introduce the background normalization as a completely free-floating parameter (μ_{bkg}). The maximum likelihood estimator will dynamically determine the true background yield by anchoring the data-driven shape templates to the high-statistics, background-dominated bins (low MLP scores) within the final discriminator distribution. This ensures that the final result is seamlessly driven by the experimental data itself, superseding the preliminary sideband estimation described in this section.

7 Systematic Uncertainties

With the adoption of a purely data-driven methodology for the background estimation, systematic uncertainties originating from hadronic interaction models and cosmic-ray composition are inherently bypassed. Consequently, the systematic uncertainties in this analysis are strictly categorized into two decoupled sources: the extrapolation of the data-driven background shape, and the signal acceptance efficiency derived from the Monte Carlo (MC) simulation.

7.1 Background Shape Uncertainty

As established in Section 6, the multidimensional distributions of the background variables exhibit robust stability as the selection approaches the Signal Region (SR). We operate under the highly confident assumption that this shape invariance persists into the SR.

However, to rigorously account for any residual, sub-dominant correlations between the input features and $R_{\mu e}$ during the asymptotic extrapolation, we construct a shape-based systematic envelope. We define the dataset at $-5.0 < R_{\mu e} < -2.85$ as the nominal background template. To evaluate the asymptotic variation, we extract alternative templates using adjacent cuts ranging from $-5.0 < R_{\mu e} < -2.65$ to $-5.0 < R_{\mu e} < -2.80$, ensuring sufficient statistical precision.

By calculating the maximum bin-by-bin deviations of these variations relative to the nominal template, and applying a mirror symmetry (upward and downward fluctuations), we construct a conservative systematic envelope. A Gaussian smoothing algorithm is subsequently applied to mitigate statistical fluctuations in individual bins. The resulting pre-fit MLP score distribution, enveloped by this shape uncertainty, is presented in Figure 14. This envelope dynamically covers potential shape biases and is treated as a continuous morphing parameter (`HistoSys`) in the final likelihood fit.

The training loss function is shown in Figure 15, there is no over training, while AUC value could reach up to 0.951 which demonstrates excellent separation power.

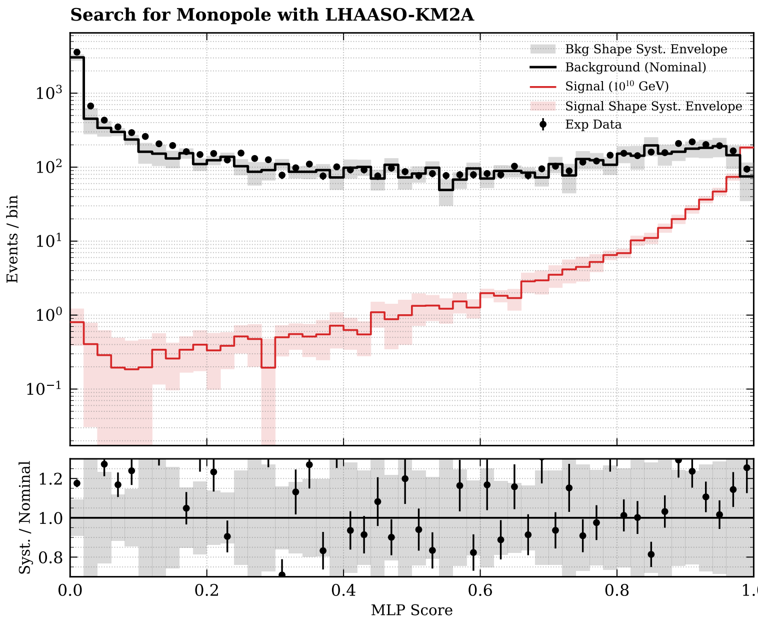


Figure 14: **Pre-fit MLP score distribution with background shape systematics.** The nominal background template (black line) is shown with its symmetric systematic uncertainty envelope (grey shaded band), derived from the asymptotic shape variations across different $R_{\mu e}$ thresholds in the control region. The signal distribution (red line) is overlaid for reference.

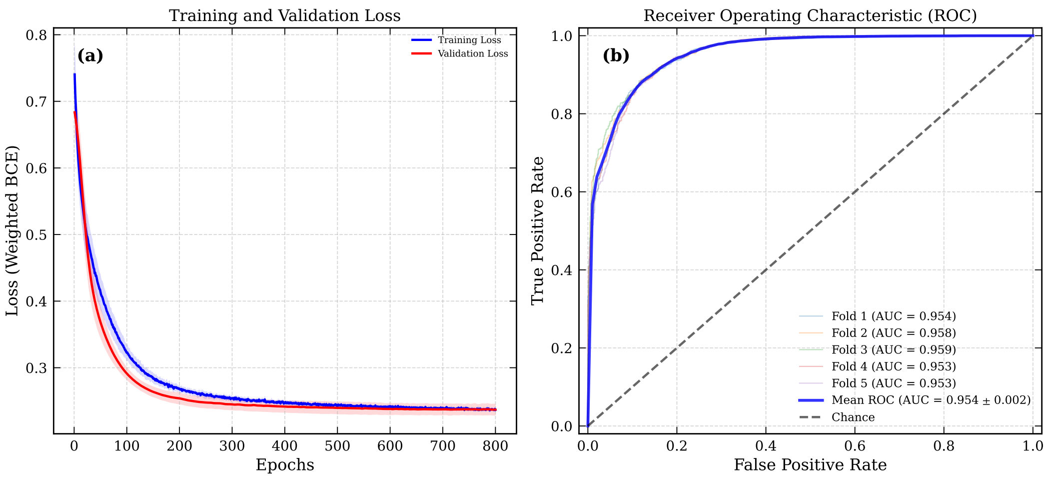


Figure 15: **Performance of the MLP classifier using 5-fold cross-validation.** (a) The evolution of Training (blue) and Validation (red) loss over 800 epochs. The shading indicates the standard deviation across the 5 folds. The consistent convergence indicates no overfitting. (b) The ROC curves for the 5 folds. The high mean AUC of 0.951 demonstrates excellent discrimination power.

7.2 Signal Simulation Uncertainties

Because the monopole signal is exclusively modeled via MC simulations, any discrepancies in the simulated detector response and shower development—primarily stemming from the choice of the underlying hadronic interaction generator—can induce systematic shifts in the signal selection efficiency and the classifier distribution. We evaluate these effects as follows:

- **Hadronic Interaction Model Dependence (Selection Efficiency):** Signal simulation uncertainties primarily originate from the choice of the hadronic interaction model, which dictates the simulated shower development. To quantify this model dependence, the selection efficiencies of the combined E_{rec} and $R_{\mu e}$ cuts defining the SR are cross-evaluated using alternative generators, including EPOS-LHC [23] and SIBYLL [24]. The maximum spread among these hadronic models is assigned as the normalization uncertainty on the signal efficiency, yielding 4.1%, 9.3%, and 1.8% for $\gamma = 10^5$, 10^6, and 10^7, respectively. The corresponding value for the benchmark point is propagated directly to the likelihood fit as a log-normal constraint.

- **Zenith Angle Migration (Spill-in Effect):** The signal MC is generated with a zenith angle up to $\theta = 70°$, matching the analysis fiducial cut. Consequently, simulated events do not account for genuine physical events with true $\theta > 70°$ that might reconstruct inside the fiducial volume ($\theta_{rec} < 70°$). Because the detector exposure and atmospheric transmission drop precipitously at high zenith angles, this pollution is intrinsically suppressed. We conservatively bound this spill-in fraction by utilizing the zenith angular resolution ($\delta\theta$). The maximum ratio of events migrating across the boundary yields an uncertainty contribution of merely 3.1×10^{-4} ($\approx 0.03\%$).

- **Core Location Migration:** Signal events are simulated with random core impacts over an expansive area ($R \leq 1000$ m). Because the analysis enforces a much stricter radial fiducial cut (typically $R \lesssim 600$ m, well within the array boundary), the probability of events outside the 1000 m simulation radius successfully triggering and reconstructing inside the tight fiducial core is vanishingly small and thus ignored.

- **Hadronic Model Shape Systematics:** Beyond the normalization effects, variations in the modeled shower longitudinal development between different generators may also induce subtle deformations in the signal MLP score distribution. To account for this, systematic shape variations are parameterized using the maximum bin-by-bin discrepancy between the alternative models (EPOS-LHC and SIBYLL) to construct "Up" and "Down" systematic templates. These shape variations are incorporated into the likelihood fit using the `HistoSys` morphing technique, ensuring that any model-dependent migration of signal events across MLP bins is correctly propagated to the final limit.

7.3 Summary of Systematics for $\gamma = 10^5$

Table 5: **Summary of Systematic Uncertainties for $\gamma = 10^5$.** The table details the sources, constraint types, and their estimated magnitude or treatment method in the final likelihood fit.

Source	Type	Magnitude / Variation	Treatment in Fit
Background Estimation			
Bkg Normalization (μ_{bkg})	Norm.	Free Floating	Unconstrained
Bkg Shape Extrapolation	Shape	Max deviation ($R_{\mu e}$ cut slices)	Interpolation (`HistoSys`)
Signal Simulation			
Hadronic Model (SR Selection)	Norm.	$\pm 4.10\%$	Log-Normal
Zenith Angle Migration	Norm.	$< 0.03\%$	Log-Normal
Core Spatial Migration	Norm.	$\sim 0\%$	*Negligible*
Hadronic Model Shape Envelope	Shape	Max bin-by-bin deviation	Interpolation (`HistoSys`)

Table 5 summarizes the systematic uncertainties incorporated into the final statistical limit calculation for the $\gamma = 10^5$ benchmark point. Because the overall signal uncertainties are exceptionally small, the analysis is heavily dominated by the available statistics and the robustness of the data-driven background shape envelope.

8 Other Lorentz Phase Space

8.1 $\gamma = 10^6$

Signal Region Definition: To effectively isolate the potential magnetic monopole signal for the $\gamma = 10^6$ kinematic regime, we optimize the Signal Region (SR) within the two-dimensional phase space of the reconstructed energy and the muon-to-electron ratio. Based on the signal Monte Carlo distribution, the SR is strictly bounded by $2.85 \leq \log_{10} E_{rec} \leq 3.10$ and $-5.00 \leq R_{\mu e} \leq -2.60$. As illustrated in Figure 16, this selection maximizes the signal acceptance while explicitly vetoing the unphysical zero-muon artifacts in the extreme deep tail (-5.00 and below).

Unblinding: Following the validation of the data-driven background methodology, we proceed to unblind the experimental data within this specific SR. Initial consistency checks are performed on the individual kinematic and morphological variables. As shown in Figure 17, the normalized distributions of the unblinded data demonstrate excellent shape agreement with the background expectations derived from the adjacent control regions, confirming the stability of the shower topologies.

Classification: The final event classification is evaluated using the dedicated MLP discriminator trained for this specific phase space. In Figure 18, the MLP output score

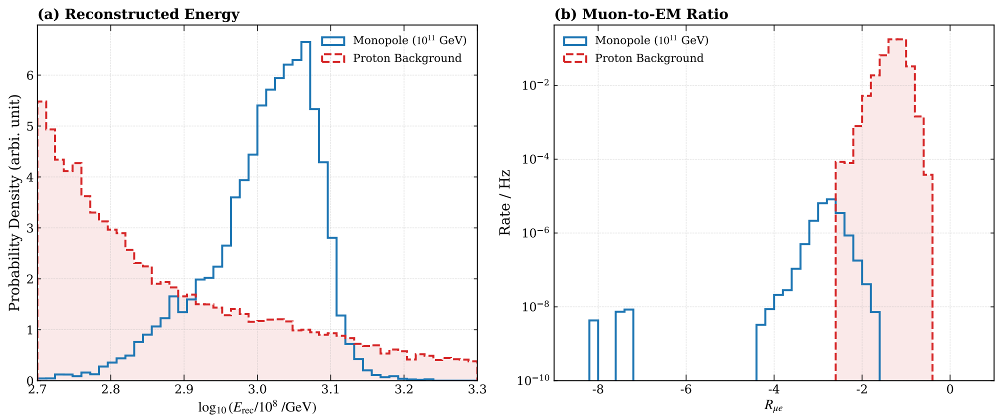


Figure 16: **Signal Region definition for** $\gamma = 10^6$**.** The optimized SR is defined by $2.85 \le \log_{10} E_{rec} \le 3.10$ and $-5.00 \le R_{\mu e} \le -2.60$, effectively isolating the signal-like phase space.

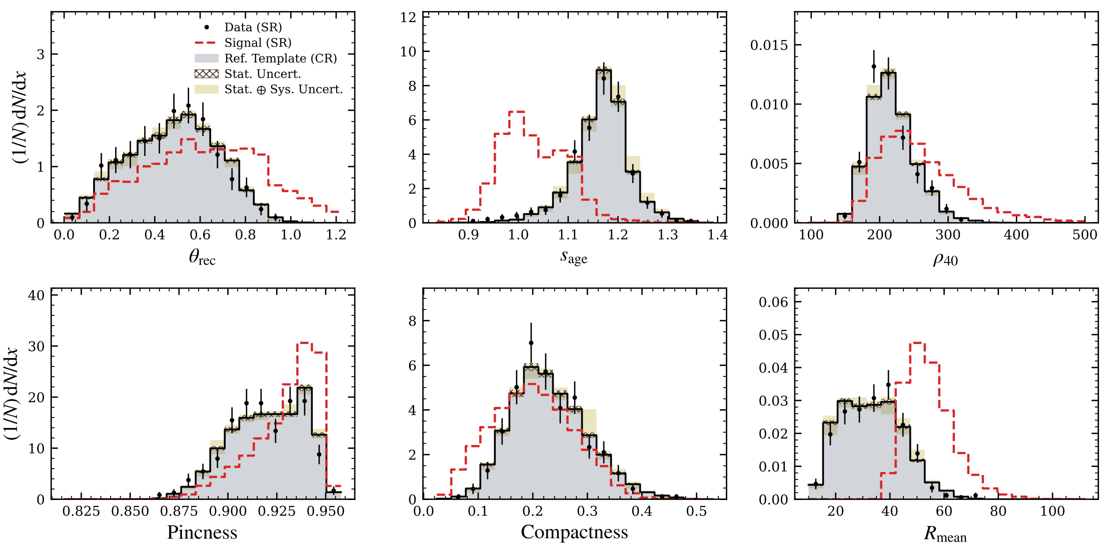


Figure 17: **Variable distributions after unblinding for** $\gamma = 10^6$**.** The unblinded data exhibits consistent topological behaviors when compared to the data-driven background templates.

distribution of the unblinded data is overlaid onto the data-driven background template, which has been appropriately scaled to the expected background yield. The observed experimental data exhibits excellent consistency with the background-only hypothesis across the entire discriminator range, revealing no significant excess in the highly-scored signal region.

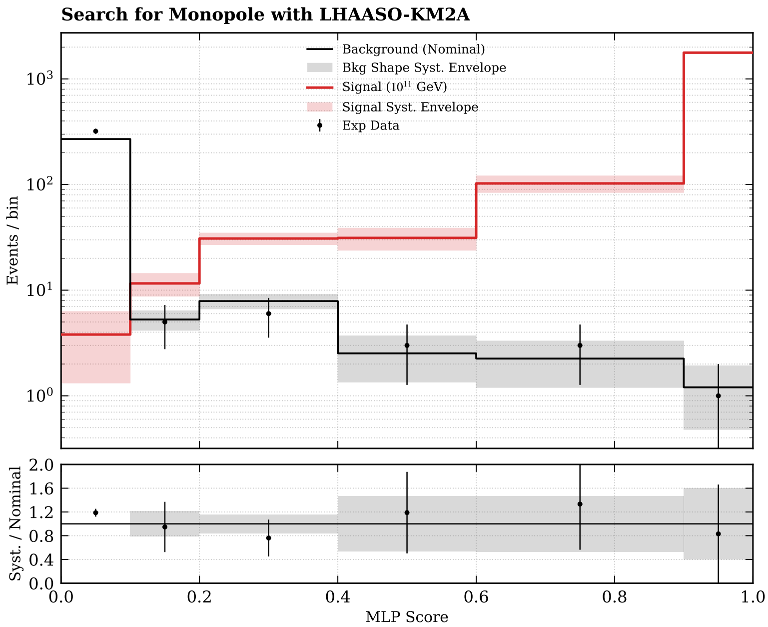


Figure 18: **MLP classification score distribution for $\gamma = 10^6$ with unequal width bin.** The unblinded data (points) aligns perfectly with the expected background template (solid line) scaled to the expected yield, indicating no signal excess.

8.2 $\gamma = 10^7$

Signal Region Definition: For the search targeting the extreme ultra-relativistic phase space ($\gamma = 10^7$), a dedicated Signal Region is established to accommodate the distinct shower characteristics at higher primary energies. The optimization yields a defined SR constrained by $3.75 \leq \log_{10} E_{rec} \leq 4.20$ and $-5.00 \leq R_{\mu e} \leq -2.20$. This expanded $R_{\mu e}$ acceptance accurately reflects the physical evolution of the monopole signal at extreme energies, ensuring high signal efficiency while maintaining robust background rejection.

Unblinding: Upon unblinding the high-energy SR, the robustness of the background model is once again confirmed. The multi-variable distributions of the unblinded events closely track the data-driven background expectations without any anomalous deviations, ensuring the input features for the neural network remain reliable.

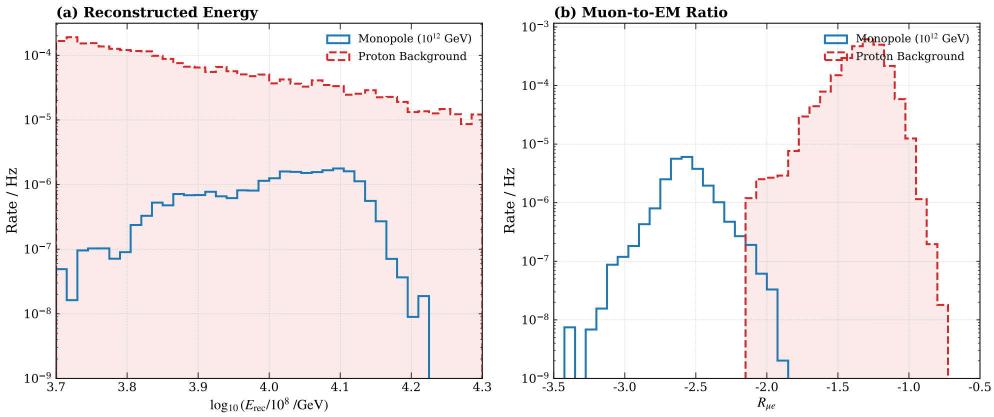


Figure 19: **Signal Region definition for $\gamma = 10^7$.** The SR is adjusted to higher energies and a slightly broader muon-poor tail ($3.75 \leq \log_{10} E_{rec} \leq 4.20$, $-5.00 \leq R_{\mu e} \leq -2.20$) to capture the ultra-relativistic signature.

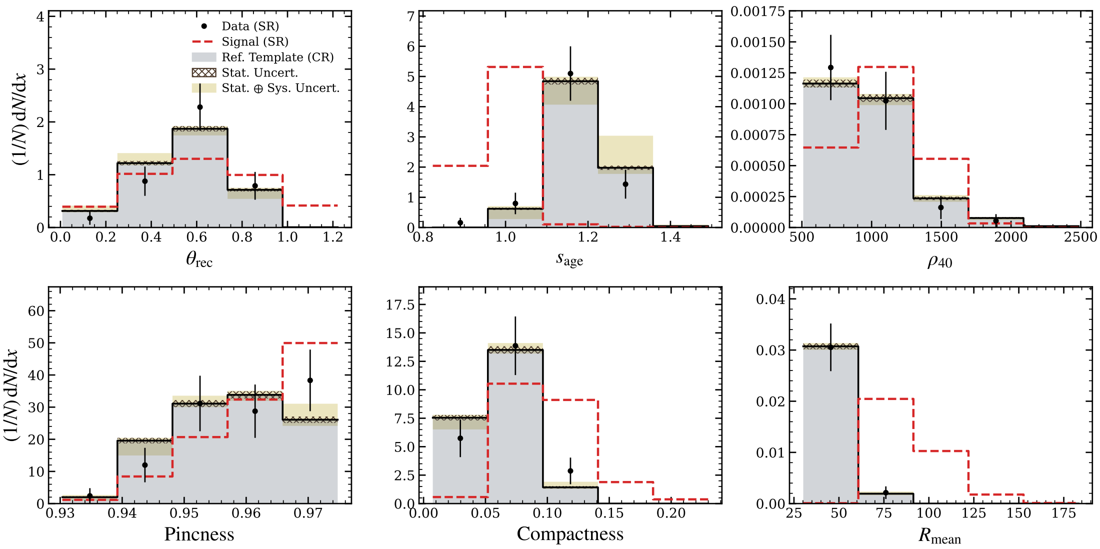


Figure 20: **Variable distributions after unblinding for $\gamma = 10^7$.** The data continues to show exceptional agreement with the background model at higher reconstructed energies.

Classification: Crucially, the final analysis is performed by evaluating the MLP score distribution for this energy tier. As depicted in Figure 21, both the total experimental yield and the shape of the MLP distribution perfectly align with the data-driven background template. This perfect alignment firmly corroborates the absence of a magnetic monopole signal in this highly energetic classification category, setting the stage for the final upper limit calculations.

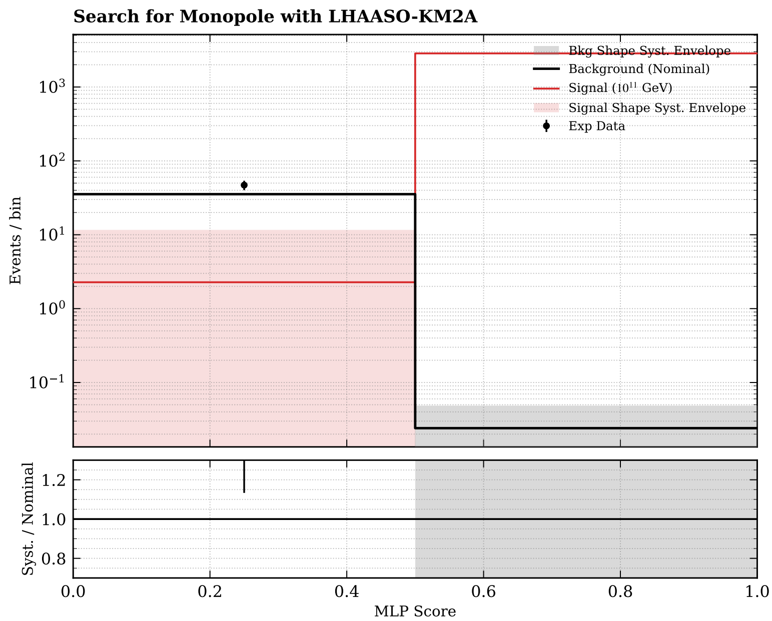


Figure 21: **MLP classification score distribution for $\gamma = 10^7$ with unequal width bin** The unblinded data shows no anomalous deviation from the background-only hypothesis, confirming the null result in this phase space.

9 Results

9.1 Statistical Framework

The statistical inference is performed using a binned likelihood fit across the final discriminator (MLP) score distribution. The analysis is implemented within the `RooStats` framework[25, 26], which constructs a parameterized likelihood function $\mathcal{L}$. To explicitly incorporate the data-driven background methodology and the systematic uncertainties evaluated in Section 7, the likelihood function is formulated as:

$$\mathcal{L}(\mu, \mu_{bkg}, \boldsymbol{\theta}) = \prod_{i=1}^{N_{\rm bins}} \text{Pois}(n_i|\nu_i) \times \mathcal{G}(\theta_{\rm shape}|0,1) \times \prod_k \mathcal{LN}(\theta_{{\rm sig},k}|\theta_k^0, \sigma_k) \times \mathcal{G}(\theta_{\rm shape}|0,1) \quad (8)$$

where the total expected number of events in the i-th bin, ν_i, is a superposition of the signal and background expectations:

$$\nu_i = \mu \cdot s_i(\boldsymbol{\theta}_{\text{sig}}) + \mu_{bkg} \cdot b_i(\theta_{\text{shape}}) \tag{9}$$

Here, μ represents the signal strength parameter (defined as the ratio of the fitted signal flux to the nominal injected flux), and $\boldsymbol{\theta}$ represents the vector of nuisance parameters. The components of the likelihood encode the physical model as follows:

- **Poisson Term:** $\text{Pois}(n_i|\nu_i) = (\nu_i^{n_i}/n_i!)e^{-\nu_i}$ describes the statistical probability of observing n_i events given the expectation ν_i in each bin.

- **Free Background Normalization:** The parameter μ_{bkg} is treated as an unconstrained, free-floating multiplier for the background expectation b_i. This ensures the total background yield is entirely data-driven and determined dynamically by the fit in the high-statistics control region (low MLP scores).

- **Background Shape Constraint:** $\mathcal{G}(\theta_{\text{shape}}|0, 1)$ is a standard Gaussian penalty term that governs the template morphing (`HistoSys`). The nuisance parameter θ_{shape} continuously interpolates between the nominal data-driven background shape and its systematic envelopes.

- **Signal Efficiency Constraints:** $\mathcal{LN}(\theta_{\text{sig},k}|\theta_k^0, \sigma_k)$ are log-normal probability density functions applied as penalty terms for the signal efficiency systematics (i.e., the E_{rec} and $R_{\mu e}$ threshold shifts). The log-normal distribution is strictly utilized here to prevent unphysical negative event yields while appropriately modeling multiplicative efficiency errors.

- **Systematic Shape Constraints**: The Gaussian penalty terms $\mathcal{G}(\theta_{\text{shape}}|0, 1)$ govern the template morphing (`HistoSys`) for both the data-driven background and the signal MC. This allows the fit to interpolate between the nominal distributions and their respective systematic envelopes, capturing potential shape distortions in the discriminator output.

To derive the upper limits, we employ the CL$_s$ method[27], a modified frequentist approach designed to mitigate over-optimistic exclusions in regimes of low sensitivity. The test statistic is the profile likelihood ratio:

$$q_\mu = -2 \ln \frac{\mathcal{L}(\mu, \hat{\hat{\boldsymbol{\theta}}}(\mu))}{\mathcal{L}(\hat{\mu}, \hat{\boldsymbol{\theta}})} \tag{10}$$

where $\hat{\mu}$ and $\hat{\boldsymbol{\theta}}$ denote the unconditional maximum likelihood estimators (MLE), and $\hat{\hat{\boldsymbol{\theta}}}(\mu)$ denotes the conditional MLE for a fixed signal strength μ. The CL$_s$ value is calculated as the ratio of the p-values for the signal-plus-background hypothesis $(s + b)$ and the background-only hypothesis (b):

$$\text{CL}_s = \frac{P(q_\mu \geq q_{\mu,\text{obs}}|s + b)}{P(q_\mu \geq q_{\mu,\text{obs}}|b)} \tag{11}$$

A signal hypothesis is excluded at the 90% confidence level (C.L.) if $\mathrm{CL}_s \leq 0.1$.

The method for evaluating the probability distributions of the test statistic depends heavily on the available statistics in the relevant phase space:

- **High-Statistics Regime (Asymptotic Approximation, for $\gamma = 10^5, 10^6$):** In regions where the expected background event count is sufficiently large, the distributions of q_μ converge to well-defined analytical forms according to Wilks' and Wald's theorems. In such cases, asymptotic approximation formulae [28] are utilized. This analytical approach provides highly accurate p-values while significantly reducing the computational overhead.

- **Low-Statistics Regime (Toy MC Method, for $\gamma = 10^6, 10^7$):** Conversely, in the deep signal-like region where the background expectation is extremely small (approaching zero), the large-sample asymptotic assumptions break down as the underlying Poisson fluctuations deviate significantly from a Gaussian limit. Under these low-statistics conditions, a fully frequentist calculation is necessitated. This involves generating a large ensemble of pseudo-experiments (toy Monte Carlo datasets) under both the b-only and $s+b$ hypotheses. The test statistic is evaluated for each toy to empirically construct the true probability distributions, ensuring a robust and conservative limit calculation.

9.2 Upper Limits on Monopole Flux

Following the statistical framework described above, we compute the 90% Confidence Level (C.L.) upper limits on the magnetic monopole flux for three distinct Lorentz factors. For each kinematic point, mass is fixed at $M = 10^5$ GeV. The results are detailed in the following sub-sections.

These limits can be directly translated into physical flux limits (Φ_{UL}) via the relation:

$$\Phi_{UL} = \mu_{UL}^{90\%} \times \Phi_{ref} \tag{12}$$

In this analysis, the reference flux Φ_{ref} corresponds to the **Parker Bound**[17], which is the theoretical upper limit on the monopole flux imposed by the survival of the Galactic magnetic field ($\Phi_{Parker} = 10^{-15}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$).

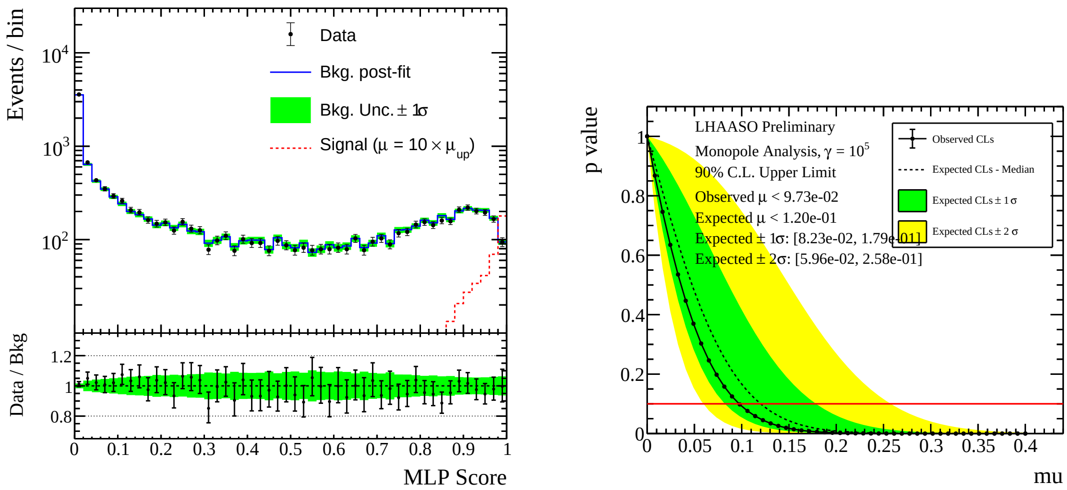
9.2.1 $\gamma = 10^5$ Search

After unblinding the signal region, no significant excess above the expected background is observed. The test statistic is evaluated to derive the exclusion limit. The numerical results for the 90% C.L. upper flux limit ($\Phi_{\mathrm{up}}^{90\%}$) are:

- **Observed Limit:** $\Phi_{\mathrm{obs}}^{90\%} = 973.2 \times 10^{-19}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$

- **Expected Limit:** $\Phi_{\mathrm{exp}}^{90\%} = 1201.5\,^{+585.1}_{-379.0} \times 10^{-19}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$ ($\pm 1\sigma$)

The corresponding CL_s scan and the systematic uncertainty bands (Brazil plot) for $\gamma = 10^5$ are presented in Figure 22a.



(a) Post-fit score distribution

(b) 90% C.L. upper limit

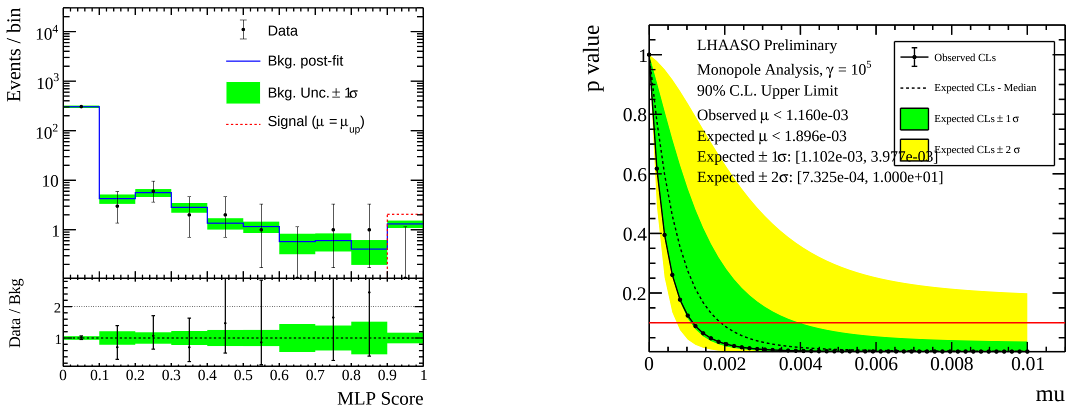
Figure 22: **Statistical inference results for the $\gamma = 10^5$ search.** (a) Post-fit classifier score distribution under the background-only hypothesis. The experimental data (black points) is compared to the background fit (solid blue line) with $\pm 1\sigma$ (green) systematic uncertainty bands. A scaled signal model (dashed red line, scaled by $\mu_{\text{up}} \times 10$) is superimposed to illustrate the expected signal shape in the high-score region. The lower panel shows the Data/Bkg ratio. (b) The 90% C.L. upper limit on the signal strength μ. The observed CL_s curve (solid black line) is compared to the median expected CL_s (dashed black line) alongside its $\pm 1\sigma$ and $\pm 2\sigma$ expected fluctuation bands. The final exclusion limit is defined at the intersection with $\alpha = 0.1$ (red horizontal line).

9.2.2 $\gamma = 10^6$ Search

Increasing the Lorentz factor to $\gamma = 10^6$, the analysis reveals excellent agreement between the data and the background-only hypothesis. The derived 90% C.L. upper limits are:

- **Observed Limit:** $\Phi^{90\%}_{\text{obs}} = 11.6 \times 10^{-19}\,\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}$
- **Expected Limit:** $\Phi^{90\%}_{\text{exp}} = 19.0\,^{+20.8}_{-7.9} \times 10^{-19}\,\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}\,(\pm 1\sigma)$

Figure 23 illustrates the profile likelihood ratio scan for this specific phase space.



(a) Post-fit score distribution

(b) 90% C.L. upper limit

Figure 23: **Statistical inference results for the $\gamma = 10^6$ search.** (a) Post-fit classifier score distribution under the background-only hypothesis. (b) The 90% C.L. upper limit on the signal strength μ.

9.2.3 $\gamma = 10^7$ Search

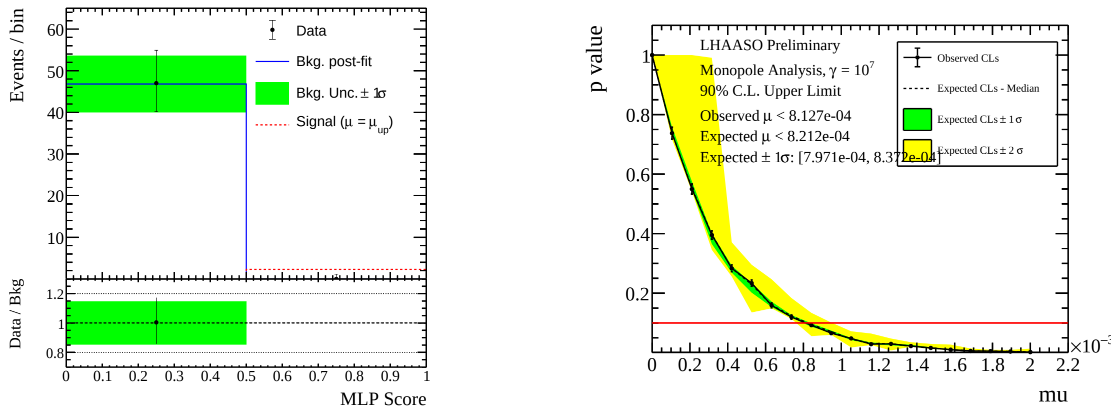
For the extreme ultra-relativistic case ($\gamma = 10^7$), the detector exhibits its highest discrimination power. The results remain consistent with the null hypothesis, yielding the most stringent limits in this study:

- **Observed Limit:** $\Phi^{90\%}_{\text{obs}} = 8.1 \times 10^{-19}\,\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}$
- **Expected Limit:** $\Phi^{90\%}_{\text{exp}} = 8.2\,^{+0.2}_{-0.2} \times 10^{-19}\,\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}\,(\pm 1\sigma)$

The limit-setting results and expected fluctuation bands are visualized in Figure 24.

9.2.4 Summary and Physical Interpretation

The comprehensive results of the 90% C.L. upper limits across all investigated Lorentz factors are summarized in Table 6. In all cases, the observed limits lie comfortably within the 1σ expected fluctuation bands, corroborating the validity of the background model.



(a) Post-fit score distribution

(b) 90% C.L. upper limit

Figure 24: **Statistical inference results for the $\gamma = 10^7$ search.** (a) Post-fit classifier score distribution under the background-only hypothesis. (b) The 90% C.L. upper limit on the signal strength μ.

Table 6: **Expected and Observed 90% C.L. Upper Limits on the Magnetic Monopole Flux.** The limits are calculated for an isotropic monopole flux with a primary energy of $E = 10^{10}$ GeV across three distinct Lorentz factors.

$\log_{10}(\gamma)$	**Expected Limit**	**Observed Limit**
	(90% C.L., $\times 10^{-19}\,\mathrm{cm^{-2}s^{-1}sr^{-1}}$)	
5	$1201.5\,^{+585.1}_{-379.0}$	973.2
6	$19.0\,^{+20.8}_{-7.9}$	11.6
7	$8.2\,^{+0.2}_{-0.2}$	8.1

A strong kinematic dependence is observed in the array's sensitivity. For highly ultra-relativistic monopoles ($\gamma = 10^7$), LHAASO-KM2A establishes an extremely stringent flux limit. However, as the Lorentz factor decreases towards $\gamma = 10^5$, the shower development characteristics begin to mimic standard cosmic-ray air showers more closely. This morphological similarity results in a reduced signal selection efficiency for the MLP discriminator, leading to a correspondingly weaker, yet still highly competitive, upper limit of $\sim 10^{-16}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$. This trend perfectly aligns with the physical expectation of the array's response to exotic particle showers.

9.3 Comparison with Existing Experimental Constraints

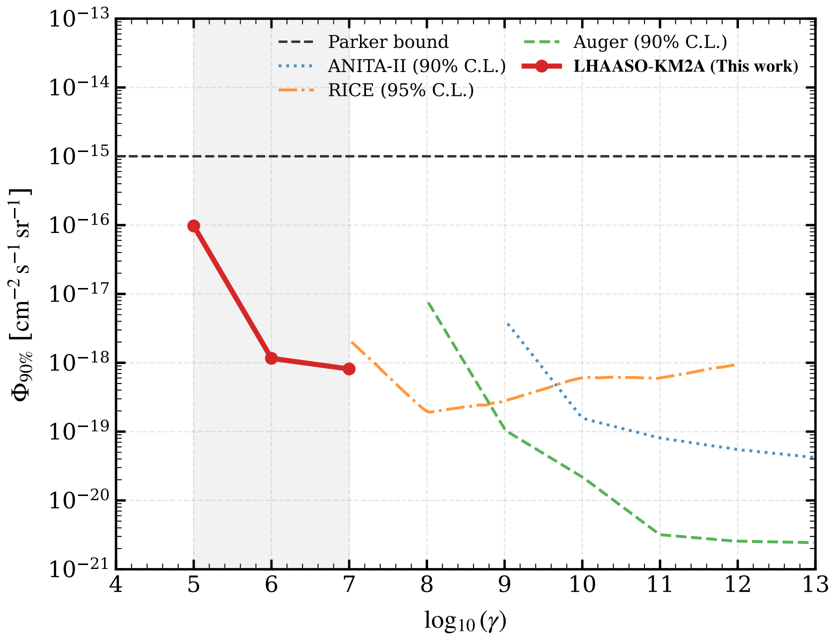


Figure 25: **Contextualization of the LHAASO-KM2A monopole flux upper limits.** The 90% C.L. upper limits derived in this work (red solid line) are compared against the astrophysical Parker bound[17] (black dashed line) and other major experimental constraints, including Auger (green)[9], RICE (orange)[11], and ANITA-II (blue)[10]. LHAASO-KM2A effectively bridges the sensitivity gap in the ultra-relativistic phase space ($10^5 \leq \gamma \leq 10^7$).

To contextualize the scientific impact of this analysis, it is essential to compare the derived LHAASO-KM2A limits with the broader landscape of magnetic monopole searches. Figure 25 illustrates the current 90% C.L. upper limits on the isotropic monopole flux as a function of the Lorentz factor γ.

Historically, the search for magnetic monopoles has been bifurcated into two distinct kinematic regimes, leaving a significant phase space gap:

- **The Sub-Relativistic Regime ($\gamma < 15$):** Deep underground or ice-cube detectors, such as MACRO [7] and IceCube [8], provide excellent and highly stringent constraints. However, their detection mechanisms (e.g., specific ionization thresholds, slow-particle trigger algorithms) render them fundamentally insensitive to highly boosted, ultra-relativistic monopoles.
- **The Extreme Ultra-Relativistic Regime ($\gamma \geq 10^7$):** Massive surface and radio arrays, including Pierre Auger [9], RICE [11], and ANITA-II [10], deploy vast effective areas to search for UHECRs. These observatories require extreme cascade energies to trigger their arrays, making them effectively blind to monopole interactions below $\gamma \sim 10^7$.

As demonstrated in Figure 25, the LHAASO-KM2A analysis perfectly bridges this uncharted territory. By leveraging the array's dense muon coverage and exceptional sensitivity to extensive air showers around the TeV–PeV scale, we establish the world's most rigorous constraints in the intermediate ultra-relativistic gap ($\gamma \in [10^5, 10^7]$).

At $\gamma = 10^7$, the KM2A limit strictly rivals and extends the lower bounds of the Auger and RICE constraints. Moving to lower Lorentz factors ($\gamma = 10^6, 10^5$), where UHECR arrays lose sensitivity, LHAASO sets unprecedented limits. Furthermore, across this entire kinematic window, the derived limits are situated ~ 1 to 3 orders of magnitude below the theoretical astrophysical Parker bound ($\Phi \sim 10^{-15}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$), placing highly meaningful constraints on the survivability of cosmic magnetic fields in the presence of such exotic particles.

10 Conclusion

In this note, we have presented a comprehensive search for ultra-relativistic magnetic monopoles using data collected by the LHAASO-KM2A detector. The analysis focused on primary mass of $M = 10^5$ GeV, exploring the critical kinematic window defined by Lorentz factors $\gamma \in [10^5, 10^7]$.

To overcome the severe depletion of Monte Carlo statistics in the extreme muon-poor phase space, we developed and validated a purely data-driven background estimation methodology. By exploiting the shape invariance of shower parameters deep in the control region, we constructed highly reliable background templates. A Multi-Layer Perceptron (MLP) neural network was subsequently trained on this hybrid data-simulation sample to maximize the discrimination power between the exotic monopole signatures and the standard cosmic-ray background.

Upon unblinding the predefined signal regions, the experimental data exhibited excellent agreement with the background-only hypothesis across all investigated phase spaces. Consequently, no evidence for a magnetic monopole signal was observed. Using a modified frequentist CL_s approach equipped with template morphing to handle systematic shape uncertainties, we derived 90% C.L. upper limits on the isotropic monopole flux.

These limits range from $1.0 \times 10^{-16}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$ at $\gamma = 10^5$ to a highly stringent $8.2 \times 10^{-19}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$ at $\gamma = 10^7$. This work successfully bridges the historical phase space gap between sub-relativistic underground searches and extreme-UHECR observatories, establishing the most rigorous experimental constraints to date in this intermediate ultra-relativistic regime.

This internal analysis note establishes the technical foundation and demonstrates the robustness of the methodology. Following the formal review by the LHAASO collaboration, these results will be finalized for official journal submission, significantly contributing to the global landscape of exotic particle searches.

References

[1] P. A. M. Dirac. Quantised singularities in the electromagnetic field. *Proceedings of the Royal Society of London. Series A*, 133:60–72, 1931.

[2] G. 't Hooft. Magnetic monopoles in unified gauge theories. *Nuclear Physics B*, 79:276–284, 1974.

[3] A. M. Polyakov. Particle spectrum in quantum field theory. *JETP Letters*, 20:194–195, 1974.

[4] John Preskill. Magnetic monopoles. *Annual Review of Nuclear and Particle Science*, 34:461–530, 1984.

[5] T. W. B. Kibble, G. Lazarides, and Q. Shafi. Walls bounded by strings. *Physical Review D*, 26(2):435–439, 1982.

[6] George Lazarides, Rinku Maji, and Qaisar Shafi. Quantum tunneling in the early universe: Stable magnetic monopoles from metastable cosmic strings. *arXiv preprint*, 2024.

[7] M. Ambrosio et al. Final results of magnetic monopole searches with the macro experiment. *European Physical Journal C*, 25:511–522, 2002.

[8] M. G. Aartsen et al. Search for relativistic magnetic monopoles with eight years of icecube data. *Phys. Rev. Lett.*, 117:011101, 2016.

[9] A. Aab et al. Search for ultrarelativistic magnetic monopoles with the pierre auger observatory. *Phys. Rev. D*, 94:082002, 2016.

[10] M. Detrixhe et al. Ultra-high-energy cosmic-ray fraction upper limits from the anita-ii experiment. *Phys. Rev. D*, 83:023513, 2011.

[11] D. P. Hogan et al. New radio limits on fast magnetic monopoles. *Phys. Rev. D*, 78:075031, 2008.

[12] S. D. Wick, T. W. Kephart, T. J. Weiler, and P. L. Biermann. Signatures for a cosmic flux of magnetic monopoles. *Astroparticle Physics*, 18:663–693, 2003.

[13] Zhen Cao et al. Lhaaso-km2a detector simulation using geant4. *Radiation Detection Technology and Methods*, 8(3):1437–1447, 2024.

[14] D. Heck, J. Knapp, J. N. Capdevielle, G. Schatz, and T. Thouw. Corsika: A monte carlo code to simulate extensive air showers. Technical Report FZKA-6019, Forschungszentrum Karlsruhe GmbH, 1998.

[15] T. Pierog, R. Engel, D. Heck, S. Ostapchenko, and K. Werner. Latest results from the air shower simulation programs corsika and conex. In *Proceedings of the 30th International Cosmic Ray Conference (ICRC 2007)*, volume 4, pages 625–628, 2008.

[16] S. Agostinelli et al. Geant4–a simulation toolkit. *Nucl. Instrum. Meth. A*, 506:250–303, 2003.

[17] E. N. Parker. The origin of magnetic fields. *Astrophys. J.*, 160:383, 1970.

[18] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[19] Koichi Kamata and Jun Nishimura. The lateral and the angular structure functions of electron showers. *Progress of Theoretical Physics Supplement*, 6:93–155, 1958.

[20] Zhen Cao et al. The first lhaaso catalog of gamma-ray sources. *The Astrophysical Journal Supplement Series*, 271(1):25, 2024.

[21] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

[22] Mervyn Stone. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):111–133, 1974.

[23] T. Pierog, Iu. Karpenko, J. M. Katzy, S. Yatsenko, and K. Werner. EPOS LHC: Test of collective hadronization with data from the LHC. *Phys. Rev. C*, 92:034906, 2015.

[24] Felix Riehn, Ralph Engel, Anatoli Fedynitch, Thomas K. Gaisser, and Todor Stanev. Hadronic interaction model Sibyll 2.3d and extensive air showers. *Phys. Rev. D*, 102:063002, 2020.

[25] L. Moneta et al. The roostats project. In *13th International Workshop on Advanced Computing and Analysis Techniques in Physics Research*, volume ACAT2010, page 057, 2010.

[26] Wouter Verkerke and David P. Kirkby. The roofit toolkit for data modeling. *eConf*, C0303241:MOLT007, 2003.

[27] Alexander L. Read. Presentation of search results: the cl_s technique. *Journal of Physics G: Nuclear and Particle Physics*, 28(10):2693, 2002.

[28] Glen Cowan, Kyle Cranmer, Eilam Gross, and Ofer Vitells. Asymptotic formulae for likelihood-based tests of new physics. *The European Physical Journal C*, 71(2):1554, 2011.